

# **Construction Guide: Data+AI Enabled Next-Gen Data Intelligence Platform**





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# 01.

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## Data challenges in the GenAI era: core bottlenecks in enterprise transformation

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Data serves as the key strategic asset for enterprises in the digital era. The emergence of generative AI (GenAI) and large language models (LLMs) is profoundly transforming enterprise productivity frameworks, collaboration models, business processes, and organizational structures. This advancement drives the transformation of enterprises from "departmental isolation" to "cross-functional collaboration" and from "hierarchical management" to "networked collaboration."

As LLMs become increasingly commoditized and the ecosystem grows more fragmented, enhancing data management capabilities to empower business value has emerged as the only sustainable competitive differentiator for enterprises. The open availability and low inference costs of models such as DeepSeek R1 have accelerated this trend, shifting the competitive focus from "model competition" to "competition for high-value data assets." The core bottleneck in AI development is increasingly becoming data quality rather than algorithms themselves. Enterprises urgently need to build systematic data engineering capabilities to unlock the potential of AI through continuous data iteration and optimization, rather than frequent model adjustments. According to Gartner's 2024 report "The Evolution of Data Management as a Discipline", over 75% of enterprises indicate that AI-ready data will rank among their top three investment priorities within the next two to three years.

The key differentiator of your organization's data is not about how much data you have, but how much high-value data you have, which is difficult for other companies to replicate. A company holding valuable data but failing to utilize it in AI and analytics initiatives is essentially welcoming disruption.<sup>1</sup>

## However, traditional data platforms are facing Common challenges in meeting the new data demands brought by GenAI:

### ► How to inherently fragmented data and maximize the value of unstructured data

Unstructured data constitutes 70% to 90% of the data in organizations today.<sup>2</sup>Fueled by generative AI initiatives, explosive growth in multimodal data processing demands, and compliance pressures, enterprises are experiencing dramatically surging needs for unstructured data management. Consequently, spending on unstructured data processing will claim a growing share of total data management expenditures.

By 2027, the IT spending focused on multistructured data management will account for 40% of total IT spending on data management technologies and services.<sup>3</sup>

Traditional data processing tools lack sufficient capabilities in metadata extraction, intelligent chunking, summary generation, and efficient vectorized storage and struggle to meet the complex requirements of GenAI applications. Therefore, upgrading the toolchain and methodology is a prerequisite for unlocking the value of massive unstructured data.

To address these challenges, Tencent has launched TCLake, an open and unified data lake service, and Elasticsearch Service (ES), which provides hybrid search capabilities. To address the challenges of massive unstructured and multimodal data in private deployment scenarios, Tencent has launched Tencent Big Data Suite (TBDS) multimodal data lakehouse service.

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<sup>1</sup>Gartner Inc., Data Management Is the Sole Differentiator in a Commoditized and Multipolar LLM World,G00828308

<sup>2</sup>Gartner Inc., Develop Unstructured Data Management Capabilities to Support GenAI-Ready Data,G00821728

<sup>3</sup>Gartner Inc., Emerging Tech: Data Fabrics With Multimodal Data Focus for Generative AI-Enabled Applications,G00818141

## ► How to break down data isolation barriers and achieve efficient transformation from data to intelligence

The true value of AI lies in transforming the vast data assets of enterprises into actionable intelligence advice. However, research indicates that the average implementation cycle for enterprise AI projects spans weeks or even months, significantly lagging behind the rapid iteration pace of business demands. The core bottleneck lies in the segregation between data management and AI development in traditional models. The workbenches of data teams and the training and deployment platforms of AI engineers are not interoperable, which creates a significant collaboration gap. This leads to:

**Overlong timeliness:** The pipeline from data to analytical prediction involves multiple systems, teams, and processes, resulting in sluggish response. For example, retail sales forecasting causes inventory issues due to long cycles.

**Unstable performance:** Top-performing laboratory models often fail to adapt to real-world business scenarios when data environments shift without timely updates of models, ultimately sacrificing decision quality.

**High expenditure due to overlapping investments:** Data engineers expend significant effort on cross-platform data transfer, format conversions, and system integration, rather than developing differentiated AI applications.

To address these challenges, Tencent has launched an integrated Data + AI solution: DLC + WeData DataOps + WeData MLOps, which combines big data, AI, and machine learning capabilities into a unified platform to deliver a better user experience and lower costs.

## ► How to address new challenges in high coherence and real-time processing of massive data

With the accelerated integration of GenAI into enterprise operations, massive, real-time, and dynamic data processing demands continue to emerge. Whether it is chat generation in customer service, financial risk control (such as credit card fraud and instant loan approval), or dynamic early warnings in supply chains, all pose challenges to system throughput, real-time processing, and horizontal scaling.

However, traditional data architectures designed for batch processing struggle to deliver second-level responses and dynamic queries for streaming data. In addition, the interaction between data and AI imposes more stringent requirements on real-time data read/write and processing latencies.

To address these challenges, Tencent has launched Stream Compute Service, which deeply integrates stream processing with AI to ensure real-time business performance.

## ► How to construct a trustworthy data governance and security system to protect the privacy and fulfill compliance

In GenAI application scenarios, data quality, compliance, and security safeguards have emerged as core challenges for enterprises. On one hand, enterprises face the dual challenges of exponential growth in data sources and severe data silos (according to statistics, enterprises manage an average of over 400 heterogeneous data sources). On the other hand, enterprises must establish dynamic and traceable data governance systems to prevent "hallucinated outputs" and mitigate privacy risks.

According to the 2023 Gartner Analytics and AI Adoption Survey, addressing risks and governance issues, along with a lack of trust in discovered analytics/metrics are top challenges for self-service analytics developers. According to the 2024 Gartner Data and Analytics Governance Survey, almost half of the respondents identify "difficulty in standardizing data across different departments/BUs" as among their organizations' top D&A governance-related challenges.<sup>4</sup>



Meanwhile, in the aforementioned 2024 webinar, when respondents were asked reasons to not leverage GenAI within ABI platforms, 28% highlighted GenAI features as not being approved by internal security/legal/compliance teams (internal policies) and another 20% mentioned lack of confidence in data quality.<sup>5</sup>

However, traditional data platforms still face many shortcomings in data governance: lack of unified data standards and business definitions; reliance on static rules, which makes it challenging to flexibly address dynamic and changing data access requirements in GenAI scenarios (such as real-time masking of sensitive information based on context); and low automation and inefficiency in governance processes. Enterprises require more intelligent and dynamic data governance and privacy protection mechanisms.

To address these challenges, Tencent has launched WeData Unity Semantics and WeData DataOps to solve issues such as data silos, business-technology gaps, and governance difficulties.

## ► How to empower all employees with data intelligence to accelerate operational efficiency

The era of GenAI requires enterprises to significantly enhance the speed of efficiency improvement and insight acquisition from data. For example, product-oriented enterprises need to interpret customer feedback in a more real-time manner, while supply chain enterprises must be able to predict logistics risks more promptly. This requires three groups of people to quickly improve their data capabilities:

**Data engineers must enhance process efficiency:** Traditional coding approaches require manual integration of SQL and Python toolchains, and the deployment of a single data analysis pipeline typically takes several days. In the future, AI-augmented data integration tools will reduce manual effort in pipeline design, error remediation, and data mapping will be greatly reduced.

**Ops personnel need to enhance operation efficiency:** Traditionally, large enterprises needed multiple Ops teams with over five years of experience each. However, with tool-driven efficiency improvements, both the headcount and knowledge requirements for Ops teams can be significantly reduced.

**Business personnel require barrier-free insight access:** Frontline personnel, unable to directly extract data using natural language (for example, "please list products with a return rate greater than 5% last month"), have to rely on IT teams for intermediary processing, resulting in delayed business decision-making. Natural language will emerge as the standard interface for data ecosystem interactions, leveraging semantic abstraction to mask underlying technical complexities; this enables significant efficiency gains in business users' data consumption.

However, traditional platforms still trap developers in inefficient and repetitive engineering cycles while forcing business personnel to endure week-long response cycles for data requests. This results in organizations struggling to unlock data value and lagging behind competitors in market responsiveness.

To enable enterprise-wide data intelligence, Tencent Cloud has launched a series of intelligent products. Among them, WeDataAgent serves data engineers and data scientists and TCInsight serves Ops personnel. To help business personnel and executives gain insights more efficiently, Tencent has introduced ChatBI, which focuses on data parsing, and TCDataAgent, which is designed for exploratory analysis, thereby driving enterprise transformation.

Therefore, the core data challenges in the era of GenAI lie in unlocking the value of unstructured data, breaking barriers between raw data and actionable intelligence, managing the processing demands of massive real-time data, establishing a trustworthy data governance and protection system, and achieving universal access to data capabilities for all employees.

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<sup>4</sup>Gartner Inc., Market Guide for Agentic Analytics,G00824122

<sup>5</sup>Gartner Inc., Quick Answer: How Is Generative AI Transforming Self-Service Analytics? G00794295

# 02.

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**Core drivers for enterprises  
to build Data+AI platforms**

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Data assets are the "fuel" of the AI era, while Data + AI platforms serve as the "refining engines" that propel data value transformation. High-quality training data determines the performance ceiling of LLMs. To prevail in this AI revolution, enterprises must build integrated Data + AI platforms, which is the core foundation for intelligent decision-making and business innovation. Typically, enterprises are driven to build Data + AI platforms by both proactive selection and reactive response:

**Proactive selection: This approach primarily targets market leading enterprises seeking to build AI-powered big data platforms that break through current boundaries and create new AI application scenarios. It includes the following two parts:**

► **Reduce "non-essential costs" of unified platforms and data foundations**

Capabilities such as LLM fine-tuning, multimodal data processing, and prompt engineering require highly specialized talents. The value of AI depends on transforming technical feasibility into business adaptability by specialized talents, such as reconstructing general LLMs into industry-specific models. The unified Data + AI platform can reduce redundant investments: Use of the standardized tech stack of the enterprise-level Data + AI platform can prevent various departments from duplicating efforts in building data pipelines and training foundational models. The platform breaks down collaboration barriers caused by single data sources, reduces internal friction caused by data silos, and lowers the communication costs of cross-departmental data alignment. Visual low-code tools enable business personnel to independently complete 80% of routine analysis and allow technical talents to focus on high-value tasks.

► **Build efficient collaboration between data and AI on a unified platform to enhance the ability to address business complexities**

Data + AI enables data and AI teams to collaborate on the same platform to implement end-to-end AI development. Data management provides efficient data support for AI applications, while AI, in turn, enhances the intelligence of data management, such as building Copilot based on LLMs. This way, a virtuous cycle is created where data and AI mutually promote and improve each other.

**Reactive response: This mainly focuses on optimizing existing data assets performance through AI enablement to address current challenges and leveraging AI to enhance team capabilities in data asset management while reducing overall costs. It includes three parts:**

## ► Enhance data governance to safeguard existing data assets

Data + AI enables more efficient governance of data, including unstructured data. This allows enterprises to avoid data security and privacy issues in AI implementation scenarios, reduce AI hallucinations, and enhance the user experience and positive feedback brought by AI usage.

## ► Improve business agility and reduce innovation waste

Executive leadership demands faster implementation of emerging technologies such as GenAI to drive business value. Currently, GPU resources are scarce in the market and extremely expensive. As a result, data team leaders must balance both agile iteration for business value and cost efficiency when they deploy Data + AI solutions.

## ► Automate data processes to improve efficiency

GenAI combined with active metadata management can automate tedious processes such as data ingestion, cleansing, and transformation and reduce the need for manual intervention. A unified data ecosystem can be used to integrate scattered data management components and simplify multi-platform maintenance.

# 03.

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**What key capabilities  
must enterprises build for  
Data + AI platforms?**

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## ► Composability of data and AI technologies

In the era of rapid technological evolution, the composability of data and AI technologies has become a core challenge for enterprises in building intelligent data platforms. Currently, swift technological advancements have shortened technology lifecycles and compressed the effectiveness window for key components. For example, in the AI field, LLMs now undergo generational upgrades, such as from GPT-3 to GPT-4, every 3–5 months. Meanwhile, technologies in data management, such as vector search and lakehouse architectures, are advancing rapidly. Technical solutions developed just six months ago may become obsolete in the near term.

Keeping existing technological stack is unlikely to maintain a core position in the long term. Take data platforms as an example. Vector databases have become a critical capability due to their support for AI semantic understanding. However, their prominence may be superseded by new storage–computing paradigms with the advancement of technologies such as graph computing and neuro–symbolic systems. In this context, enterprises need to build a "pluggable Data + AI platform" to achieve three key capabilities through a modular architecture:

### **Flexible replacement**

Components such as data storage, computing engines, and model services can be decoupled as standardized modules to support rapid replacement with technological iterations.

### **Dynamic scaling**

Emerging technologies, such as quantum computing accelerators and novel multimodal models, can be seamlessly integrated through API Gateway and microservice architectures.

### **Cost control**

The need to completely overhaul the existing architecture due to technological upgrades can be avoided. Instead, only partial upgrades are required to adapt to new technological developments, thereby preventing high costs and waste.

## ► End-to-end Data + AI lifecycle development and business integration

The Data + AI development process involves data processing, model construction, and LLM training. The platform should provide comprehensive development tools and enable end-to-end management from data to AI models to ensure the deep integration of data and AI. Meanwhile, various teams can collaborate efficiently on a unified platform to reduce management costs and improve development efficiency.

By building an integrated Data + AI platform, enterprises can dramatically shorten the data-to-AI delivery cycle (from months to weeks or less) while boosting data engineering efficiency by 50% or more. For example, in the financial industry, evolving regulatory policies demand rapid compliance reporting to avoid penalties. Customers can use the real-time data pipeline of the platform to integrate multi-source data such as transaction records and credit information, enabling rapid iteration and dynamic optimization of risk models to quickly meet compliance requirements.

## ► Processing and enhancement of multimodal data (structured and unstructured data)

AI applications typically involve using structured and unstructured data. Therefore, the Data + AI platform should support multimodal data management to enable enterprises to use diverse types of data effectively throughout the Data + AI development process. The platform should transform multimodal data into machine-interpretable, business-actionable "Intelligent Fuel" that propels enterprises from data storage to data-driven operations.

## ► Unified metadata-driven governance and compliance

To ensure high-efficiency output of AI applications, the platform must balance data accuracy, availability, and security while prioritizing model quality and actual results of applications. Therefore, the platform must provide governance capabilities for Data + AI, including metadata management, data quality, and security. By building a governance framework to unify data semantics, enterprises can differentiate the roles between data owners and data users, ensuring that the responsibilities and rights of data usage are balanced. This helps to proactively address business concerns regarding data usage during Data + AI implementation. The platform should implement a unified governance framework to achieve comprehensive management of data and AI, enhancing the overall performance and reliability of AI applications.

## ► Agentic Analytics

In the Data + AI platform, Agentic Analytics significantly enhances the value of data-driven business by using intelligent data management, automated decision-making, and proactive analysis capabilities. For example, automated data cleansing use AI agents to detect dirty data, such as missing values and outliers, and proactively perform fixes, such as imputation and interpolation. Dynamic data lineage tracking autonomously records data sources, transformation paths, and usage history to ensure compliance and traceability.

## ► Cloud-native elastic architecture and multi-cloud deployment

Enterprises adopt decoupled cloud service architectures (such as logical data warehouses and data lakes), which are combined with FinOps tools to optimize resource usage and ensure cross-cloud data compliance and cost control. Converged data management platforms further reduce the procurement of independent systems.

To build a new-generation integrated Data + AI platform, enterprises must implement technological integration (such as Data Fabric and RAG), architectural optimization (multi-cloud decoupling), and organizational transformations (skill enhancements). The purpose is to help enterprises maximize the utilization of data assets, adapt swiftly to market changes, and build continuous innovation capabilities based on AI. This is not only a technological upgrade but also a key strategy for enhancing core competitiveness and business value.

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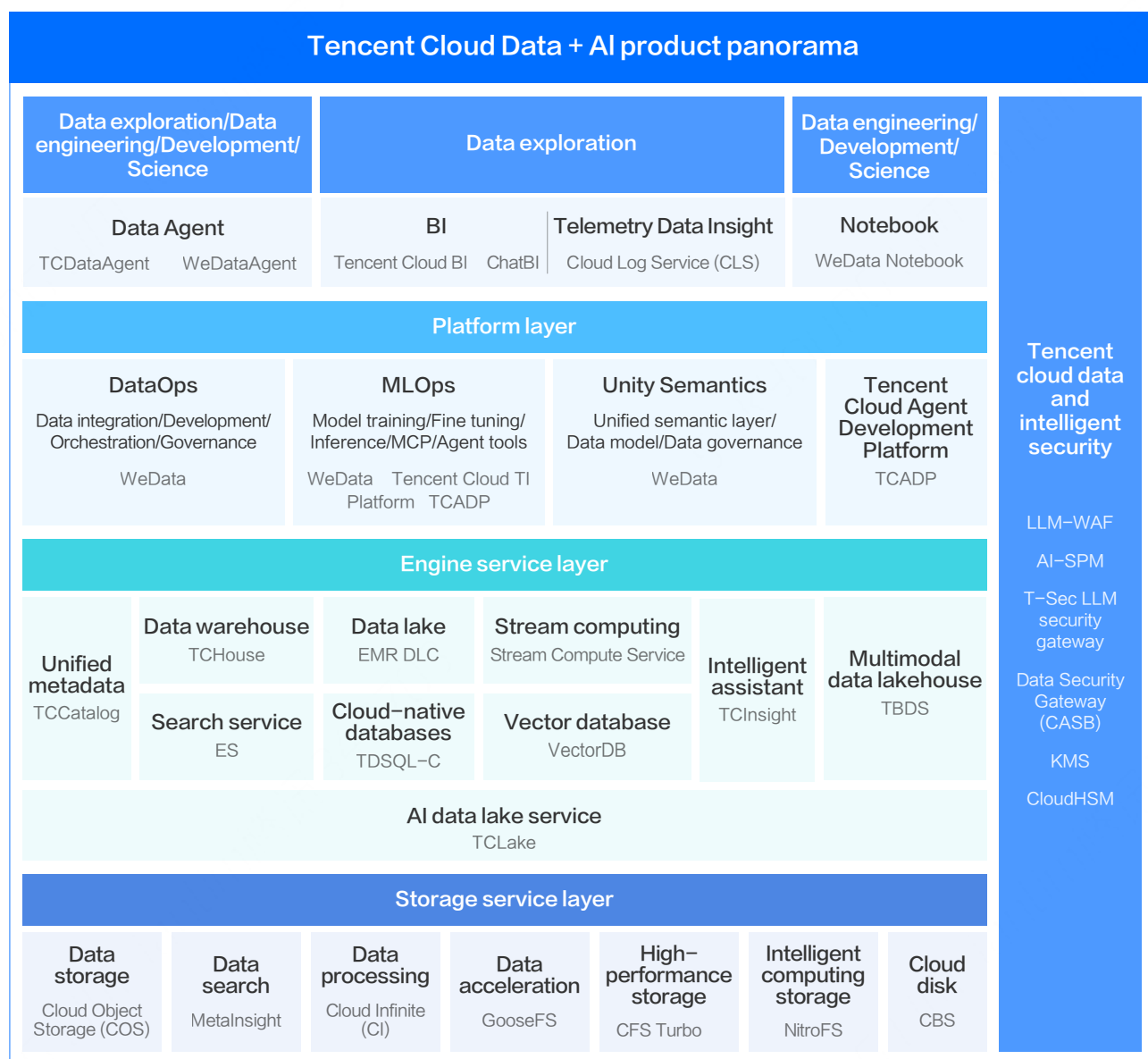
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## Product solutions and strengths of Tencent Cloud Data + AI

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Facing the opportunities and challenges of the GenAI era, Tencent Cloud is committed to building efficient and intelligent enterprise-level Data + AI infrastructure through deep integration and innovation of Data + AI technologies, thereby accelerating the digital and intelligent transformation and efficiency revolution of enterprises. The Data + AI platform of Tencent Cloud is built around data management as its core and integrates products and services such as AI computing power, data storage and analysis, data governance, security management, AI model training, and real-time decision-making. It provides an end-to-end solution from data ingestion to intelligent applications.



## Challenges of enterprises

In the global wave of digital transformation, data has transcended its role as a mere byproduct of IT systems to become a key strategic asset that drives business innovation, optimizes decision-making processes, and builds core competitive advantages. Most enterprises face the following challenges in their quest to extract insights and monetize value from massive data:



### High cost of data interpretation

The business meaning of data is unclear, and large amounts of complex data are difficult for business personnel to intuitively understand, making it even harder for AI to comprehend.



### Low efficiency in data analysis

Data query and report creation heavily rely on IT support or specialized data analysts, which not only prolongs the response cycle and struggles to adapt to rapidly changing business demands, but also leads to inefficiencies in extracting data insights.



### Lack of unified governance and efficient collaboration

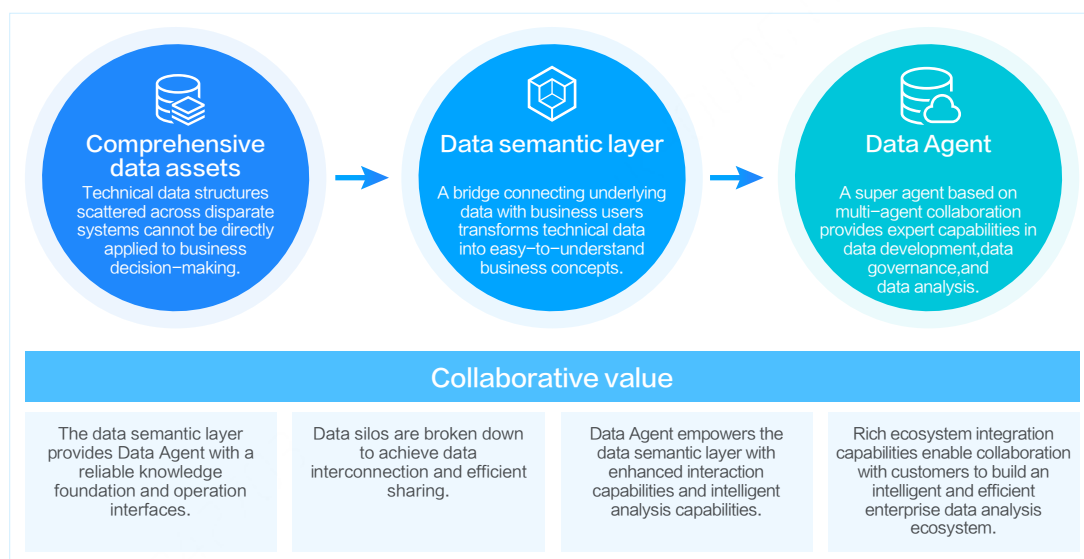
Data quality represents the foundation of analysis and decision-making. Enterprises often lack agile and user-friendly governance tools to enforce sustainable data governance practices within applications.



## Solution

WeDataAgent uses the Data + AI dual engines to provide a full-linkage data intelligence solution. At the data layer, a unified semantic layer is built based on the metric platform and governance center to achieve standardized dimensional modeling, metric definitions, and metadata management. This addresses the issues of data interpretation discrepancies and communication barriers caused by inconsistent data standards and unclear definitions, significantly improves the manageability of data assets and the clarity of business meaning, and greatly reduces the cost of data interpretation.

Leveraging the big data and LLM infrastructure of Tencent Cloud, along with its technology reserves and open architecture, WeDataAgent helps users quickly build data intelligence systems based on natural language interaction. IT departments are directly empowered with AI, allowing them to focus on developing exclusive intelligent applications that serve enterprise businesses. Business personnel can gain insights from data and quickly generate result reports without needing professional SQL knowledge. Data teams are evolving from data extraction service providers to designers and builders of enterprise data intelligence engines, significantly enhancing their influence and value while accelerating the large-scale implementation of data-driven and AI-powered applications.



At the AI layer, the following core features are used to improve data analysis efficiency and drive governance implementation, thereby gradually enhancing data quality:



### Data interpretation and insight capabilities

A Language User Interface (LUI) is built to enable natural language interactions for code generation, data Q&A, and metric analysis. This allows business personnel to obtain data insights from conversations without needing SQL knowledge, thereby achieving data democratization.



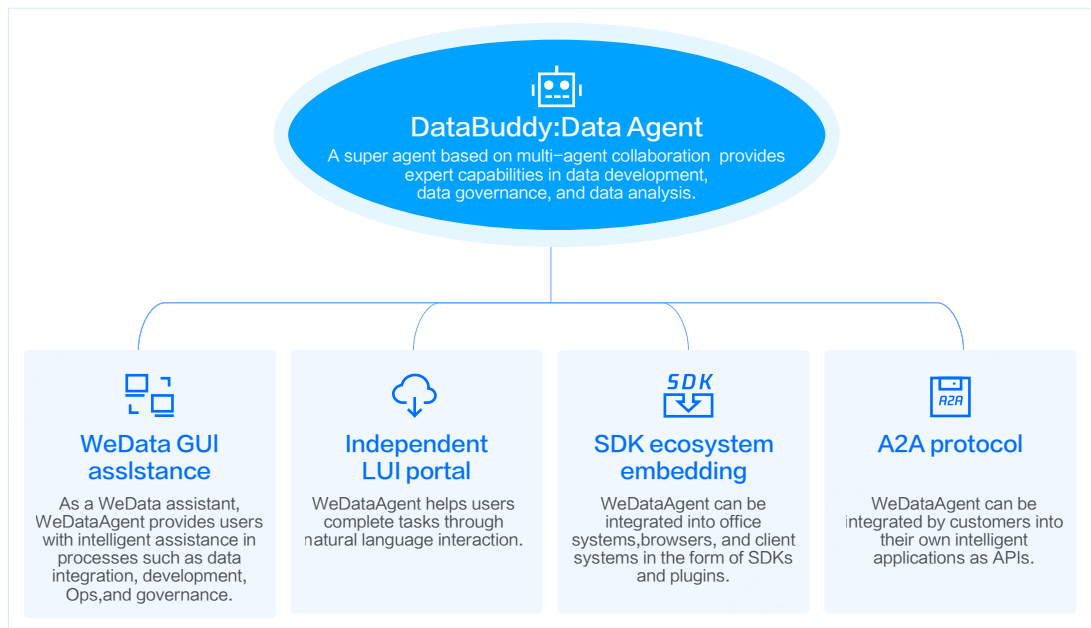
### Intelligent planning and execution capabilities

WeDataAgent provides features such as environmental awareness, intent recognition, and automatic error correction. These features enable WeDataAgent to understand complex data requirements and plan the optimal execution path. WeDataAgent can transform abstract needs into concrete actions by calling multiple tools.



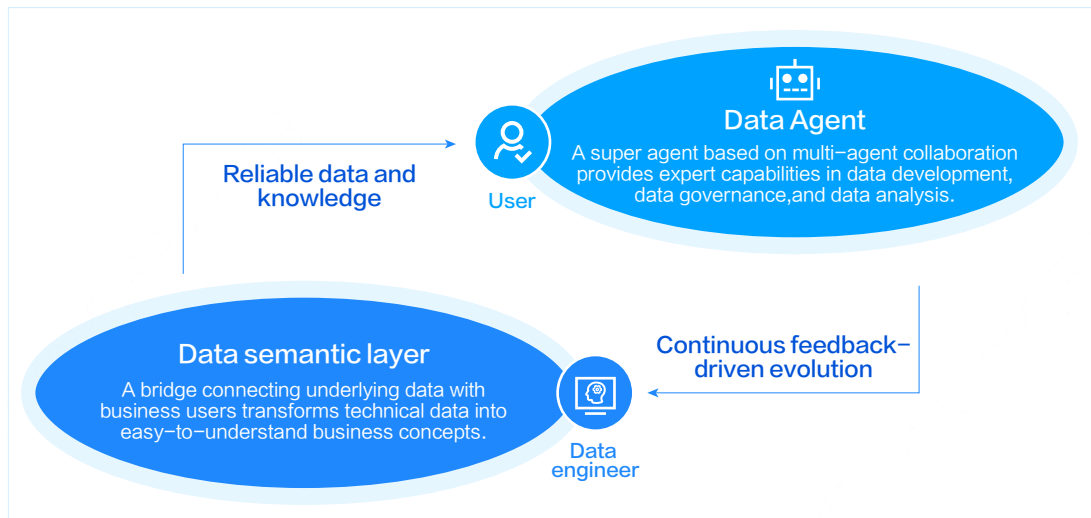
### System integration and scalability capabilities

WeDataAgent supports Model Context Protocol (MCP) and Agent-to-Agent (A2A) protocols. It can be seamlessly integrated into the business systems and self-built Agent systems of customers in the form of plug-ins and APIs/SDKs.



### Dual driver of data utilization and governance

WeDataAgent is an iteratively evolvable data agent that supports user Q&A feedback, annotation, and collaborative governance, systematically driving continuous improvement in data quality while enhancing the accuracy of agent analysis. This creates a positive feedback loop of data optimization and analysis capability enhancement.



## Use cases

### Scenario 1: intelligent decision support for marketing teams

Marketing planners can directly ask WeDataAgent through its natural language interface: "Analyze the core user profile of those who participated in the A product promotion last month, including age distribution, geographic characteristics, and main interests." WeDataAgent, relying on marketing-related metrics defined by a unified semantic layer, automatically consolidates multi-source data from Customer Relationship Management (CRM) systems, user behavior logs, and transaction systems to generate multidimensional, visual user profile reports in real time.

### Scenario 2: "agile analysis assistant" for business personnel

When product operations personnel detect abnormal fluctuations in user metrics, they can query WeDataAgent: "Analyze the possible main reasons for the recent decline in daily active users (DAU) of the application and compare the differences in the churn rate across user segments (such as new and existing users and users on different device types)." WeDataAgent then performs task orchestration, and may call predefined metric models or perform multidimensional data drilling to provide initial attribution analysis clues and data insights.

### Scenario 3: "intelligent assistant" for data governance

When business users or data analysts have questions about a specific data metric, they can query WeDataAgent: "What is the specific calculation formula for the monthly repurchase rate of the metric? What are the raw data tables and intermediate processing steps it relies on?" WeDataAgent can trace dependencies based on its metadata knowledge and data lineage, then explain the complete lifecycle of the metric to users in clear, understandable terms.

## Strengths



### Full-linkage solution

WeDataAgent is not just a simple AI-empowered Q&A tool, but a complete data intelligence platform based on a unified semantic layer. It achieves end-to-end process coverage from data governance to intelligent analysis and provides bidirectional driving capabilities.



### Potential for building an open and collaborative ecosystem

WeDataAgent adopts a modular component design and provides open APIs and SDKs to enable deep integration and collaboration with various existing enterprise information systems, such as business intelligence (BI) systems, CRM, Enterprise Resource Planning (ERP), Office Automation (OA), and industry-specific applications, thereby allowing data value to permeate all aspects of the business.



### Dual expertise support of LLMs and big data

Tencent's combined technical strengths in LLMs and enterprise-grade big data processing ensure precise interpretation and professional-grade results.

## Tencent Cloud data analysis agent TCDDataAgent

### Challenges of enterprises

With the continuous advancement of GenAI technologies, we believe that AI Agents will soon become an integral part of the enterprise workforce, freeing up valuable time for employees to focus on higher-value challenges facing the business. Data Agent is a specialized category of AI Agent that focuses on the field of data analysis. Data Agent functions as a senior data expert within an enterprise. It is familiar with the complex data structures and industry-specific background knowledge. It masters advanced data preprocessing techniques to efficiently integrate messy raw data. In addition, Data Agent can gain deep insights from the data, predict future development trends, and generate tailored data analysis reports and decision-making suggestions for the enterprise based on analysis results.

Despite continuous improvements in model quality and reductions in inference costs, enterprises commonly face the following challenges when they deploy trustworthy AI Agent systems:



### Accuracy

In enterprise applications, AI Agent outputs must meet high quality standards, particularly for critical business features like finance where near-zero error tolerance is required.



### Controlled data access

AI Agents need to be able to access various data sources to reliably operate within a business context. These data sources include unstructured data, such as text and audio, and structured data, such as tables and views, which are often distributed across multiple systems.

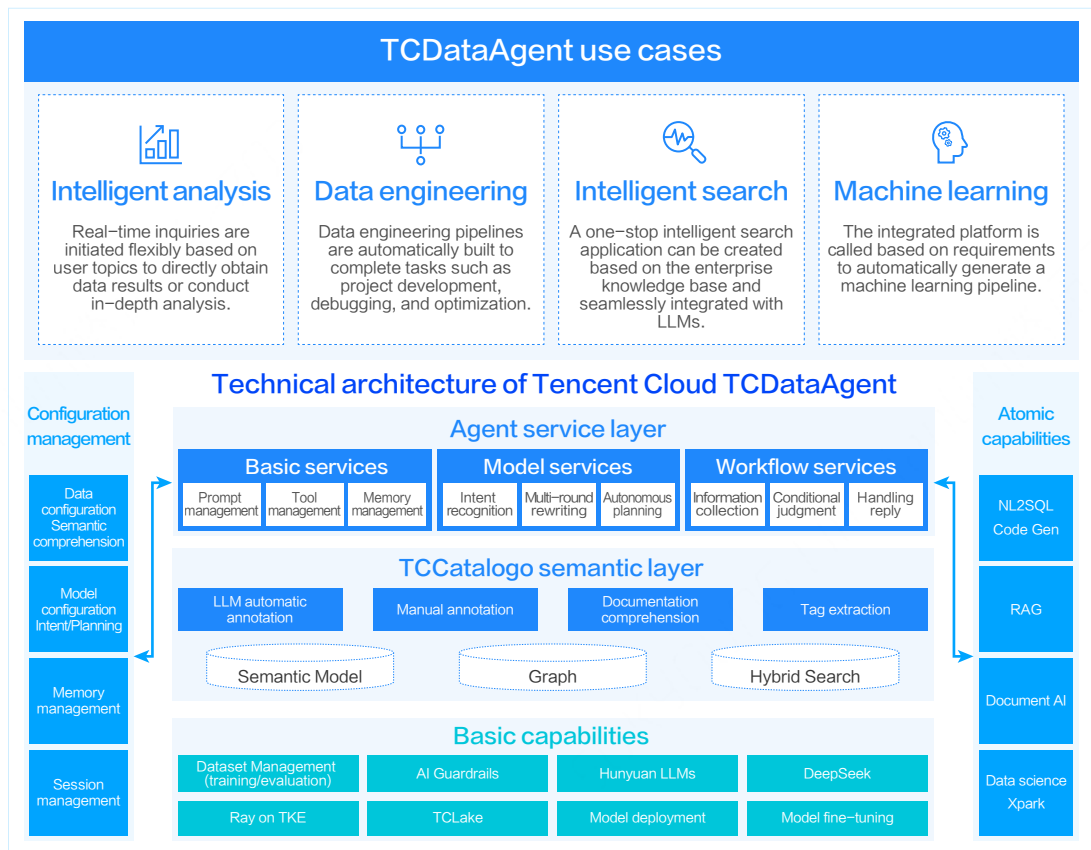
Looking ahead, a growing number of enterprises will deploy agent analytics within critical decision workflows, driving a fundamental shift from human-driven insight generation to autonomous "data-to-action" cycles. Compared with legacy BI systems, decision latency will decrease by an order of magnitude, signifying a radical transformation of corporate decision-making processes. In the past, enterprises relied on analytical tools to assist decision-making during the data-to-insight process. In the future, Data Agent will autonomously conduct data analysis and deliver actionable insights with optimization recommendations, eliminating the need for continuous human intervention.

Take the retail industry as an example. In the future, enterprises will need to only delegate inventory forecasting tasks to Data Agent, such as analyzing the sales trend of the past three months and predict the inventory requirements for the next quarter. Data Agent will automatically connect to the sales database, invoke AI models to generate predictions, and present the results in the form of visual charts. In addition, Data Agent can proactively offer suggestions, such as "Based on the trend, it is recommended to increase the inventory of product A while reducing the procurement of product B."

## Solution

To address these challenges, we launched TCDDataAgent this year, which is a fully managed agent service designed to help enterprises integrate, retrieve, and analyze both structured and unstructured data. TCDDataAgent helps users better understand data intuitively and extract valuable insights. In addition, enterprise users can easily build high-quality AI Agent applications based on TCDDataAgent.

TCDDataAgent can correctly interpret user intentions, proactively plan tasks, use tools to execute tasks, and improve responses by reflecting on the results. During task execution, TCDDataAgent leverages atomic capabilities including NL2SQL, NL2Py, AI Search, and XPark, which are combined with LLMs to analyze data and generate accurate responses. In addition, TCDDataAgent supports protocols such as MCP and A2A and enables seamless integration with third-party AI applications.

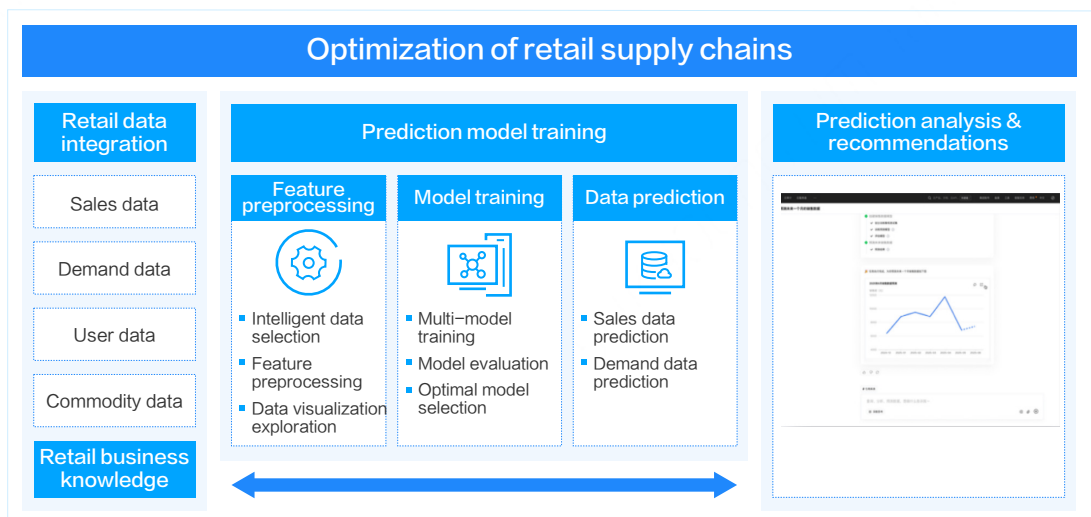


## Use cases

### Use case 1: supply chain forecasting

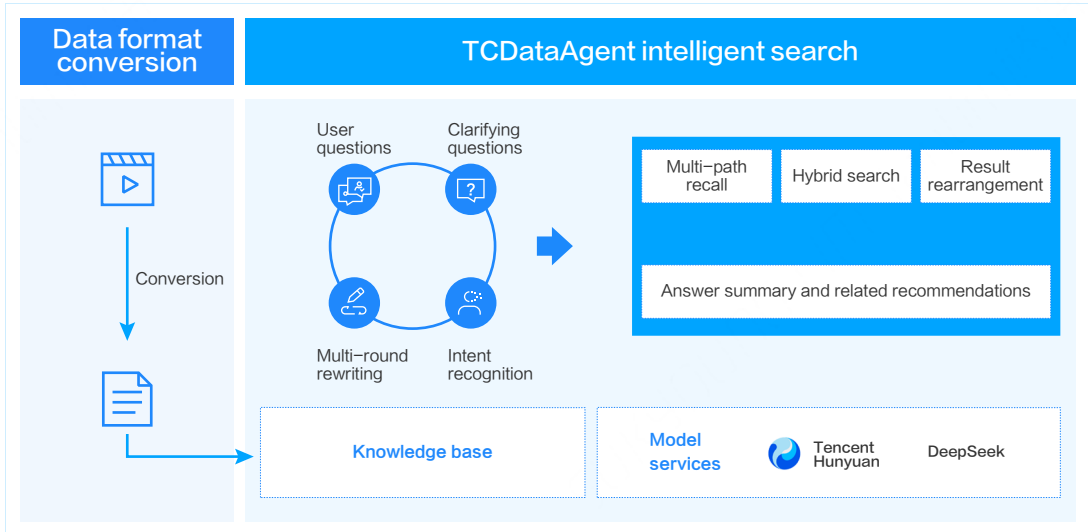
A Southeast Asian company uses AI models for demand and sales forecasting to optimize inventory and supply chain management. However, AI algorithm development and tuning are complex, model iteration cycles are long, and prediction accuracy remains suboptimal. Data deviation between sales forecasts and demand predictions frequently leads to inventory overstock or shortages.

TCDATAAGENT assists users in implementing the AutoML process, shortening prediction cycles, and continuously iterating and optimizing prediction results based on business data and scenarios. Based on forecasting results, TCDATAAGENT effectively collaborates sales data with demand data, provides recommendations to support decision-making, and promotes cost reduction and efficiency improvement for enterprises.



## Use case 2: video intelligent search

In the media industry such as television networks, a large amount of program videos have been accumulated. With the rise of mobile Internet, many people wish to intelligently search for past content and view related program videos. TCDDataAgent enables users to build video RAG systems by storing video-transcribed text in its knowledge base. Leveraging its intelligent search capabilities, TCDDataAgent retrieves relevant program clips, delivers AI-associated video responses to users, and supports interactive chats and contextual recommendations.



## Use case 3: WeChat Reading "AI Q&A"

WeRead Reading is a popular online reading platform that provides hundreds of millions of users with a large number of books, comics, WeChat official accounts, and online listening services. ES supports intelligent search for book content. The platform can have a complete understanding and cognition of search words, which supports open-ended question answering and rich interaction capabilities, such as book citations and guessing what you want to ask.





## Strengths

Tencent Cloud Big Data has accumulated a wealth of experience in the field of data analysis over the years and built a series of high-quality data analysis tools available for TCDDataAgent to use, such as TCAlyst (for structured data processing) that supports natural language interactions, AI Search and Document AI (for unstructured data processing), Meson (for high-performance computing), and XPark (for unified distributed computing). With these foundations, along with the unified metadata service TCCatalog and unified lake storage system TCLake, TCDDataAgent can provide enterprises with high-quality and high-performance data analysis agent services.



### Structured data processing

Different from typical text-to-SQL systems that rely solely on pattern matching, TCAlyst uses a semantic model to map business terms to underlying data. This approach significantly improves NL2SQL accuracy in complex business scenarios involving multi-table joins.



### Unstructured data processing

ES natively provides hybrid search capabilities. It combines keyword search and vector search to deliver large-scale, high-quality, and low-latency search services for unstructured data, such as text, audio, and images. By leveraging atomic services such as embedding and rerank provided by "Intelligent Search Development," users can effortlessly build RAG frameworks to create AI-powered Q&A applications. In addition, the Document AI feature enables rapid extraction of structured data and textual content from documents.



### High-performance computing

Meson is Tencent Cloud's self-developed next-generation high-performance computing engine, featuring convergence, openness, and intelligence. It supports diverse scenarios, such as batch processing, interactive analytics, and machine learning (ML), and provides underlying high-performance computing support for TCDDataAgent.



### Unified distributed computing framework

Xpark is a high-performance distributed computing framework developed by Tencent Cloud. It is compatible with commonly used DataFrame+ML interfaces in the Python ecosystem and provides integrated capabilities for data analysis, data preprocessing, model training, and model inference. Xpark integrates with TCDDataAgent to enable natural language-driven data analysis, prediction, and decision-making support.

## Tencent Cloud BI intelligent assistant – ChatBI

### Challenges of enterprises

When enterprises conduct BI data analysis, they often need to establish dedicated data IT teams to generate reports by using customized development or agile BI tools. This process is both labor-intensive and time-consuming. This model fails to meet the data analysis needs of frontline business personnel due to insufficient IT resource support. In addition, self-service BI tools present usability challenges for non-technical users. Frontline personnel lack the methodologies of professional analysts, making it difficult for them to quickly obtain analytical conclusions. In the wave of AI, enterprises expect to leverage AI technologies to lower barriers and enhance analytical efficiency.

## Solution



### Conversational analysis

Customers can pose questions in natural language. Based on technologies such as LLM capabilities, RAG, and NL2DSL, Tencent Cloud ChatBI can accurately interpret the customer's semantics, perform data analysis, and return data conclusions. ChatBI significantly reduces the complexity of data analysis and decreases the analysis time to seconds. Moreover, customers can configure a knowledge base by inputting industry-specific terms into the LLM, allowing it to better understand industry scenarios and become an expert in a particular field.



### Data insight report

Based on LLM capabilities, ChatBI can interpret customer data, identify overall trends, detect data changes and their causes, and provide summaries and recommendations. For abnormal data fluctuations, ChatBI can further analyze the influencing factors and offer targeted business optimization suggestions to the customer. Customers can also input interpretation templates to adapt ChatBI data insights to their business scenarios and even directly generate a complete, visually enriched data interpretation report.



### BI report creation and analysis

In the BI report creation process, AI capabilities enable customers to automatically generate complete reports, interpret existing reports, and generate complete analysis reports.

## Use cases

### Use case 1: performance analysis for the retail industry

A well-known retail customer deployed ChatBI based on LLMs. In addition to being applied in financial data analysis scenarios for executives, the focus is on expanding data analysis support for front-line business personnel across nationwide stores. The customer leveraged ChatBI to significantly simplify data analysis, enabling a wide range of frontline personnel to become "analysts", while using AI technologies to identify business issues and obtain operational recommendations.

### Use case 2: business analysis in the cultural and tourism industry

A cultural and tourism state-owned enterprise, in addition to reviewing core data from fixed reports, also requires its managers to flexibly query various business data daily. Previously, this demand required a large amount of IT data manpower support. After ChatBI is deployed, it supports dozens of business scenarios for the customer. After differentiated knowledge bases are configured, ChatBI meets the data analysis needs of different managers from various perspectives. AI-powered insights and recommendations enhance operational efficiency for customers.

## Strengths



### Best practices from Tencent Cloud businesses

Tencent Cloud ChatBI evolved from real-world business applications within Tencent Cloud. It supports comprehensive business analytics across all Tencent Cloud services. From executives to frontline operational analysis employees, all use Tencent Cloud ChatBI for data analysis, accumulating extensive practical experience in performance optimization and product refinement.



### Deep optimization for vertical data analysis scenarios

Industry-specific NL2DSL/NL2SQL engines with built-in dedicated semantic parsers support customized industry-specific computing templates and can directly generate complex SQL logic. The unified query abstraction layer seamlessly adapts to multi-source heterogeneous databases such as MySQL and Hive and automatically handles field type differences.



### Enterprise-grade fine-grained permission system

A hybrid model of Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC) is adopted to achieve multi-level permission isolation and support fine-grained control at the row and column levels. Full-linkage logs can be automatically recorded and queried to meet compliance audit requirements.



### Native BI visualization capabilities

Based on Tencent Cloud BI capabilities, ChatBI supports a wide range of exquisite visual charts and provides a smooth interactive analysis experience.

## WeData Notebook

### Challenges of enterprises

Jupyter Notebook, as an integrated development environment (IDE) for interactive analysis, has become an indispensable tool in the fields of data analysis, data science, and AI. It helps users quickly explore data and build models with flexibility and visualization capabilities. However, Jupyter Notebook is used for data analysis, users often face the following challenges:

1. Users often need to perform tedious environment preparations and service configurations, including installing required libraries, configuring data connections, and allocating computing resources. This process is not only time-consuming but also prone to errors, particularly for beginners, because it involves a high level of complexity.
2. Enterprises typically build big data storage and computing engines such as data lakes and data warehouses. When they perform data analysis by using self-built Jupyter Notebook, the data retrieval efficiency is relatively low, especially in massive data scenarios, making it difficult to extract the required data.
3. In a multi-user environment within enterprises, file and resource sharing and management become exceptionally complex. Issues such as data conflicts, version control, and resource contention must be addressed during team collaboration to ensure the efficiency and effectiveness of team work.

Therefore, enterprises need to explore more efficient workflows and tools, optimize environment configurations, enhance data retrieval speeds, and ensure effective resource isolation and sharing, in order to fully leverage Jupyter Notebook and accelerate data-driven decision-making.

### Solution

To address the preceding challenges, WeData Notebook is launched to provide users with an out-of-the-box interactive analysis IDE. This innovative feature not only simplifies the data analysis process but also achieves integration of data processing, model training, and analysis workflows through seamless connection with the upstream and downstream modules of WeData, significantly enhancing work efficiency of users.



### Quick environment setup

When WeData is adopted, users do not need to manually install Python or configure environment dependencies. They can create a Notebook workspace with a few clicks, thereby enabling an out-of-the-box experience. The workspace provides built-in images such as Scikit-learn, TensorFlow, and PyTorch, which enable users to efficiently perform data preprocessing, ML model training, and predictive data analysis.



#### Seamless integration with big data engines

Notebook workspaces support binding with Tencent Cloud EMR and DLC engines to automatically retrieve cluster configuration information. Users can directly submit Notebook tasks to big data engines, reuse the computing resources of the engines, and quickly retrieve data from the engines for interactive analysis.



#### Complete resource isolation and sharing mechanism

In team collaboration scenarios, WeData provides dedicated Notebook running environments to ensure resource and configuration isolation between users. In addition, WeData supports file-level access control. Users can configure different access permissions based on their business requirements to ensure the security and privacy of sensitive data.

## Use cases

#### Financial industry: risk analysis and prediction

For a financial industry customer, risk management is one of the core tasks. In the credit approval process, Notebook is used to build a credit scoring model that assesses the likelihood of borrower default, thereby optimizing lending decisions and reducing non-performing loan (NPL) ratios.

#### Healthcare industry: disease prediction and analysis

A healthcare industry customer trains ML models by using Notebook to identify early signs of diabetes and predict the progression of the disease in patients. Based on the prediction results, personalized treatment plans are developed to improve patient outcomes and quality of life.

#### Retail industry: customer behavior analysis

For a retail industry customer, analyzing customer behavior and purchase patterns is a critical strategy for driving sales growth. The customer uses Notebook to build a recommendation system that pushes personalized product recommendations to customers, thereby increasing conversion rates and customer satisfaction. In addition, customer churn rates can be analyzed and targeted marketing campaigns can be developed to enhance customer loyalty.

## Strengths



#### Data + AI integrated processing pipeline

WeData Notebook is deeply integrated with Tencent Cloud big data engines. By incorporating comprehensive DataOps capabilities, including workflow orchestration, scheduling, and Ops, WeData Notebook enables automated data deployment from development environments to production environments. WeData Notebook seamlessly integrates with MLflow for experiment and model management, enabling a rapid and complete workflow from data preprocessing to model training and evaluation. This connects the full linkage from data development and governance to AI development and application.



#### Intelligent IDE development experience

WeData integrates Notebook tasks, data integration, and development tasks into a single platform to provide a unified IDE that shields users from the differences in underlying technical architecture. WeData Notebook is integrated deeply with the Git version control system, thereby supporting Git-based collaborative development and version management processes. WeData Notebook supports online collaboration among multiple users. Team members can share progress and feedback in real time to enhance the efficiency and quality of data science and development work.



### Rich AI ecosystem integration

WeData Notebook provides DeepSeek-based workspace templates, enabling users to rapidly launch DeepSeek AI application development. WeData Notebook is integrated with the AI-powered Tencent Cloud TI platform. Therefore, Notebook files can be submitted to the high-performance platform for training. WeData Notebook provides a built-in AI code assistant based on Tencent Cloud Hunyuan and DeepSeek. This AI code assistant helps with tasks such as code writing, code commenting, and code error correction, making data development more efficient and intelligent.

## WeData DataOps

### Challenges of enterprises

In the era of rapid Data + AI development, enterprises face challenges such as the explosion of data sources, severe data silos, and the fragmentation of development and Ops. According to statistics, enterprises typically manage over 400 heterogeneous data sources on average. Traditional data management models face issues such as long development cycles (an average of 2–3 months), difficulty in failure location, and lack of guaranteed data quality. Especially in the AI era, 80% of the time is spent on data preparation, which severely hinders the process of digital transformation and intelligent upgrading for enterprises.

DataOps introduces DevOps principles into the data field to build a comprehensive automated management system for the entire data lifecycle.

### DataOps solution



#### Agile data development

DataOps provides visual data flow orchestration, intelligent code generation, and automated testing frameworks, significantly improving development efficiency. DataOps supports visual development, code development, and visual orchestration by using rich data source connectors, and reduces the technical complexity.



#### Intelligent quality control

DataOps enables proactive identification and automatic resolution of data quality issues based on an ML-powered exception detection engine. DataOps builds a comprehensive data quality metric system and rule engine to ensure data credibility from the source.



#### Continuous integration and deployment

DataOps establishes a CI/CD pipeline for data links, enabling automated deployment from development to production environments. DataOps supports advanced release policies such as code approval and package release, which ensures the security of changes and business continuity.



#### Full-link observability

DataOps provides end-to-end data lineage tracking, real-time performance monitoring, and intelligent fault diagnosis. Predictive maintenance and intelligent tuning are achieved based on MLOps technologies to significantly reduce the fault recovery time.



### AI-native integration

DataOps natively supports capabilities such as automated feature engineering and model version management for AI development and Ops. It simplifies model development processes by using AutoML technologies and accelerates the iteration and innovation of AI applications.

By synergizing these capabilities, DataOps achieves automation, standardization, and intelligence in data development and Ops and fundamentally addresses the efficiency, quality, and cost challenges of enterprise data management.

## Use cases

### Retail e-commerce scenarios

In omnichannel marketing, DataOps enables real-time integration and intelligent analysis of customer data. It breaks down online and offline data silos, builds comprehensive customer profiles, and enables precise marketing and personalized recommendations. The end-to-end process, from data access, data development, task testing, task submission, and task deployment to business label calculation and marketing outreach, has been automated to improve marketing efficiency and customer experience.

### Financial industry practices

In risk management scenarios, DataOps supports the full-process automation from data collection and feature engineering to model deployment. By integrating multi-source data, such as transactions, credit records, and behavioral data, through real-time data pipelines, DataOps enables rapid iteration and dynamic optimization of risk models, significantly improving the timeliness and accuracy of risk identification.

## Strengths



### Flexible architecture for multi-scenario adaptation

WeData provides a comprehensive DataOps solution to address the complex and diverse data management needs of enterprises. DataOps supports cross-environment deployment for different tenant accounts, meeting the multi-level management needs of large enterprise groups. DataOps enables cross-project deployment within the same tenant environment to facilitate collaborative development between departments. DataOps provides multi-engine instance deployment within a single project to ensure hitless migration between development, testing, and production environments. These multi-level, multi-dimensional deployment capabilities fully adapt to the organizational structures and management requirements of different enterprises.



### Enterprise-level security and compliance assurance

WeData has built a comprehensive security and control system to meet the data security and compliance requirements of different industries. From tenant isolation to access control, every step strictly adheres to the best practices for data security. WeData delivers tailored security policies and compliance solutions, regardless of whether addressing stringent regulatory requirements in finance or intellectual property protection in manufacturing.



### Seamless integration of native development experience

WeData deeply integrates with the Git version control system to support Git-based collaborative development and deployment processes. Developers can use familiar development tools and workflows to perform operations such as code management, branch merging, and version rollback through Git. This native integration approach not only lowers learning barriers but also retains the established development patterns of the development team, facilitating the effective adoption of the DataOps methodology.

Based on the previously mentioned differentiated capabilities, WeData not only provides a technologically advanced DataOps platform, but more importantly, understands and meets the practical needs of enterprises during their digital transformation process, helping them build a data development and Ops system tailored to their unique characteristics.

## WeData MLOps

### Challenges of enterprises

The essence of MLOps is to address the following four challenges in enterprise business and IT, thereby achieving the cost reduction and efficiency improvement goals of large-scale AI development for enterprises:



#### Model deployment time far exceeding expectations

According to Algorithmia's 2020 report, 64% of enterprises take more than a month to deploy a new model, and 55% of ML projects remain in the experimental phase.



#### Unsustainable model services

The speed of model iteration and deployment cannot keep pace with rapidly evolving business requirements.

Models are at risk of degradation (data drift) from the moment they are launched.



#### Challenges in model lifecycle management

Assets such as code, data, algorithms, and models lack unified version management and traceability.

Enterprises lack defined standards for the ML and AI model lifecycles from production to application.



#### Challenges in cross-team collaboration

Significant discrepancies exist in tools and workflows across different teams.

The silo effects and communication gaps between the business, Ops, and AI teams are challenging to overcome.

### MLOps solution

Leveraging Tencent Cloud's foundational capabilities and its deep accumulation in the fields of data and AI, WeData will further develop the MLOps feature architecture shown in the following figure, based on the existing data platform. This will enable a seamless end-to-end process for data management, data development, model training, model processing, and service delivery, achieving efficient and comprehensive AI development lifecycle management. MLOps will also provide several key features, thereby precisely addressing the challenges in enterprise business and IT operations.



#### Manageable processes and reproducible experiments

Comprehensive version-controlled archiving is implemented for data, features, environments, and code. Strict version management is applied to data, code, configurations, models, hyperparameters, and pipelines, and dependencies are recorded. When model issues occur, rapid fault locating is performed, allowing step-by-step backward tracing from services to data to identify and resolve root causes.



### Evaluable models and controllable outcomes

MLOps implements end-to-end integration testing and supports non-functional requirement testing (security, fairness, interpretability, and responsible AI). Deployment and updates are allowed only after the corresponding standards are met.



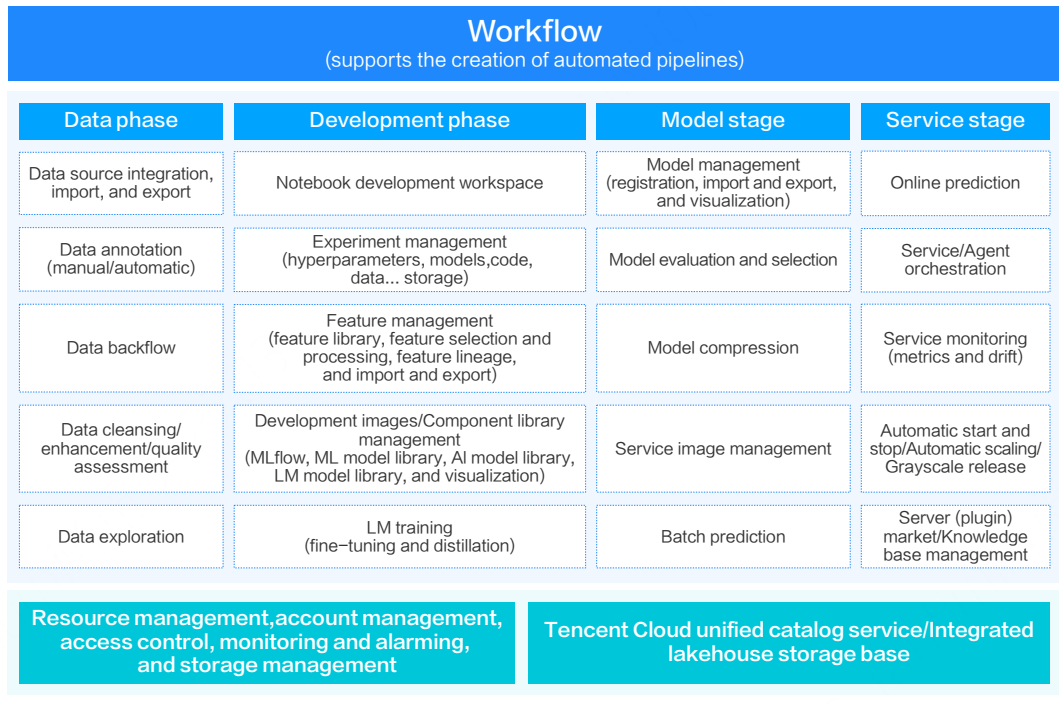
### Deployable models and monitorable services

MLOps supports cloud-based deployment and provides comprehensive monitoring capabilities, including service performance metrics, model effectiveness metrics, and data drift detection, to ensure optimal online performance.



### Automated end-to-end development

MLOps supports the creation of automated pipelines to achieve high-level automation across processes, such as data development, feature engineering, model training, evaluation and selection, service deployment, and delivery integration. This enables automated Continuous Integration/Continuous Delivery/Continuous Training (CI/CD/CT) to reduce costs in large-scale AI development.



## Use cases

### Scenario 1: continuous training and model updates

Problem: Data distribution changes over time (such as user behavior, market trends), and the models need to be dynamically updated.

Case: In e-commerce recommendation systems, real-time user click and purchase data automatically triggers model retraining to maintain recommendation accuracy.



### Scenario 2: monitoring and drift detection

Problem: Model performance degradation or changes in data distribution, such as data drift and concept drift, are detected.

Case: In a credit scoring model, the system monitors whether the distribution of input data features (such as the user's income range) has changed. If the distribution changes, an alarm or model retraining is triggered.

### Scenario 3: resource optimization and cost control

Problem: Training and inference resources are managed to balance performance and cost.

Case: Cloud services automatically scale in or out: The number of GPU instances are dynamically adjusted based on the number of inference requests.

### Scenario 4: end-to-end lifecycle management for AI products

Question: The end-to-end lifecycle from data preparation to model retirement is managed.

Case: An intelligent customer service system automatically collects user feedback data, cleanses the data, retrains models, performs grayscale releases, and fully replaces outdated models.

## Strengths

WeData MLOps of Tencent Cloud provides the following unique strengths:



WeData has developed robust and comprehensive capabilities in data storage, governance, and development. It further provides MLOps capabilities to achieve integration between data and AI. This can better help customers streamline the entire process from data to models to services, manage the dependencies between models and data, and enable more powerful MLOps capabilities.



Based on Tencent Cloud's powerful foundation, sufficient resources, stable service provision, multi-region capabilities, strong product R&D teams, and reliable delivery and maintenance capabilities, WeData MLOps enables customers to confidently migrate their business to the cloud and achieve the secondary discovery and utilization of data value.

## WeData Unity Semantics

### Challenges of enterprises



#### Data silos and integration complexities: obstacles to unified analysis

In most enterprises, data is typically scattered across isolated, disconnected business systems. A "siloed" information system architecture has led to significant differences in data formats, storage methods, interface standards, and other aspects between different systems. This creates what is known as "heterogeneous system" barriers. This situation prevents enterprises from achieving a holistic, unified data view, leaving data scattered like unstrung pearls that cannot be woven into a complete business panorama.

Data silos directly prevent enterprises from performing effective cross-domain data analysis and deriving deep value insights. Any analytics project that requires integrating multiple data sources faces a long data integration cycle, high development costs, and unpredictable risks.



### Gap between business needs and technology solutions: the "last mile" challenge in value realization

The ultimate goal of data analysis is to serve business decisions. However, significant "language" barriers often exist between technical teams and business personnel. Business personnel may find it difficult to describe their complex and evolving data requirements and underlying business logic by using accurate technical terms, while data analysts and engineers may produce data analysis results that are disconnected from actual business needs due to lack of a deep understanding of specific business scenarios, resulting in a "supply-demand mismatch."

The gap between business needs and technology solutions directly leads to delayed response to data analysis needs, making it difficult to transform analytical results into actionable business insights. IT departments are often overwhelmed by the constant influx of unclear, ad-hoc data requests, while business departments are dissatisfied due to insufficient data support. Consequently, the strategic goal of data-driven decision-making becomes increasingly elusive for enterprises.



### Challenges in data governance and consistency: unstable foundation of trusted data and insufficient reliability of data agents

Many enterprises have accumulated vast amounts of data, but the quality of the data remains less than ideal. The absence of unified data standards and business definitions leads to frequent data quality issues, such as inaccuracy, incompleteness, untimeliness, and inconsistency. Data governance systems are inherently complex, emphasizing the lifecycle management, standardization, consistency, and traceability of data. However, they are difficult to implement effectively under traditional architectures. Untrusted data hinders the effective deployment of data agents.

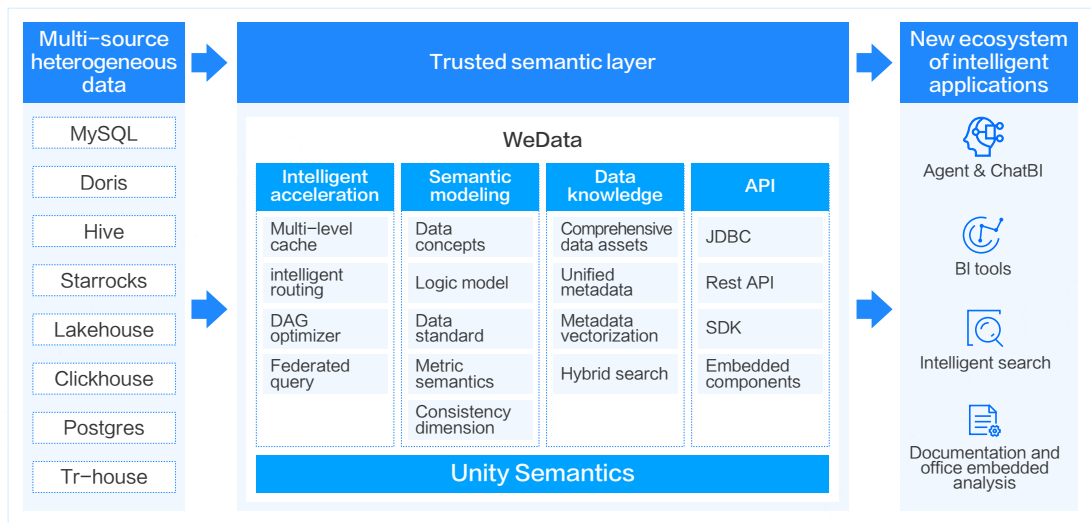


### Low development efficiency and high maintenance costs: heavy drag on agile responses

The traditional data development model, especially one that relies heavily on manually written ETL scripts, often involves lengthy processes and struggles to adapt to evolving business requirements. Data pipelines are complex and fragile. If a failure occurs at any stage, the troubleshooting and repair process is often time-consuming and labor-intensive. IT resources are heavily consumed by low-value repetitive tasks and cumbersome system maintenance, leaving inadequate capacity to focus on driving business innovation and rapid market responsiveness.

## Solution

By building a unified semantic layer, communication barriers are overcome, allowing departments such as finance, operations, and technology to interpret and analyze data using a consistent language. The unified semantic layer is not a single technical point, but a systematic solution. It reinvigorates enterprise data capabilities from three dimensions: technical architecture innovation, business value reinvention, and ecosystem building. It effectively addresses data silos, business-technology gaps, governance challenges, and efficiency bottlenecks.



### NoETL ushers in a new era of data engineering

At its core, NoETL (also known as Zero-ETL or ETL-less) leverages technologies such as data virtualization, automated metadata management, intelligent query optimization, and federated query to minimize or even eliminate the large-scale, periodic physical ETL processes in traditional data warehouses.

The unified semantic layer plays a crucial role in the NoETL architecture. It defines the data model, business objects, and their logical relationships from the perspective of business users. NoETL technologies dynamically connects, queries, and integrates underlying heterogeneous data sources across disparate physical locations based on these semantic definitions. This means that data can be accessed and used in its original location, achieving "no data migration, semantic online." This multi-pronged approach effectively addresses the issue of data silos, enabling enterprises to rapidly adapt to business changes and perform on-demand data acquisition and analysis, without waiting for long ETL development cycles.



### Headless BI achieves the separation of semantics and presentation, driving innovation in analytical applications

The essence of Headless BI is to decouple the semantic layer (the "brain" of data analysis), including data models, business logic, metric definitions, and calculation standards, from the presentation layer, including data visualization and reporting tools. In this architecture, the semantic layer becomes an independent, reusable component that provides consistent and standardized data services and metric definitions for various frontend analytical tools and applications. Its core value lies in enabling "define metrics once, reuse everywhere," eliminating the chaos of redundant metric definitions and maintaining multiple logic sets across different BI tools or applications.

The unified semantic layer is the "brain" of the Headless BI architecture. It encapsulates enterprise-wide standardized business terms, calculation methodologies for key performance indicators (KPIs), dimensional hierarchies, and data models. By using APIs and similar methods, this centralized semantic layer delivers consistent and reliable data semantic interfaces to various upper-layer applications, including traditional BI reporting tools, agile BI platforms, Jupyter Notebooks for data scientists, emerging natural language query (NLQ) applications, and even LLMs. This not only significantly enhances the efficiency and consistency of data analysis, but also provides enterprises with greater flexibility when they select or replace frontend analytical tools.



### Modern data architectures such as Data Fabric and Data Mesh are integrated

In the architectural concepts of Data Fabric and Data Mesh, the unified semantic layer plays a more critical role. In Data Fabric, the semantic layer serves as the critical enabler for making data discoverable, comprehensible, trustworthy, and interoperable, and helps to automate metadata management and data integration. In Data Mesh, while individual business domains autonomously govern their data products, they must leverage a shared, standardized semantic layer to enable cross-domain data comprehension and interoperability, ensuring a balance between domain autonomy and global consistency. The unified semantic layer provides core technical support for the implementation of these advanced data architectures.

## Use cases

### Full-linkage user growth analysis for a retail enterprise

Retail e-commerce enterprises typically operate across multiple customer touchpoints, including online channels (such as apps, websites, mini programs, and third-party e-commerce platforms) and offline stores. The unified semantic layer can integrate omnichannel user data, including browsing, clicks, searches, cart additions, orders, payments, reviews, returns, and exchanges, to build a unified user behavior metric system, including the visitor count, conversion rate, average order value, repurchase rate, and churn rate. Through these semantic metrics, enterprises can gain clear insights into user behavior patterns, preferences, pain points, and key conversion points across the entire consumer journey, thereby optimizing marketing strategies, improving product design, enhancing user experience, and driving business growth.

### An Internet technology enterprise conducts agile A/B testing to quickly validate iteration results based on data-driven insights

Optimization and feature iteration for Internet products, such as social apps, search engines, content platforms, and online games, heavily rely on A/B testing. Based on the standardized definitions and unified calculations of core business metrics, such as DAU, Day 1 Retention Ratio, Average Session Duration, Click-Through Rate (CTR), Conversion Rate (CVR), and Paid User Ratio, the unified semantic layer supports the efficient deployment of large-scale A/B testing, automated data collection, and scientific evaluation. Product and operations teams can quickly assess the effectiveness of different strategies based on consistent and trustworthy metric data and make more informed product decisions to drive agile optimization and rapid iteration.



### Ecological integration layer

Provides rich downstream ecosystem integration capabilities, including RESTful API, JDBC, GraphQL, JavaScript SDK, Python SDK, and office documentation plugins.



### Semantic modeling layer

Provides metadata management, business term definitions, dimensional modeling, and metric modeling to build a unified business semantic model.



### Adaptive acceleration layer

Provides automatic and self-service Materialization acceleration configuration based on History-Based Optimization (HBO) and Cost-Based Optimization (CBO), and supports multi-level cache acceleration based on the intelligent data source routing of user queries.



#### Data Fabric layer

Logically interweaves data across heterogeneous data sources based on a unified data catalog, thereby creating a unified enterprise data logic map to break down data silos.



#### Data source layer

Connects to various enterprise data sources, including structured data, semi-structured data, and unstructured data.



#### Build a comprehensive data asset catalog: enabling data to be discoverable, understandable, accessible, and operable

**Business-oriented enhancement of metadata management:** Traditional metadata management focuses more on the technical attributes of data, such as table structure, field types, and storage location, which is incomprehensible to business users. The unified semantic layer maps and associates these technical metadata with business concepts, such as business terms, business objects, metric definitions, and data domain divisions, "translating" pure technical descriptions into a "data language" that business users can understand and use. This is the core of building a truly business-oriented data asset catalog that serves the business. The goal of a data asset catalog is to enable data managers to identify their available data, including its sharing scope and security level.

**Deliver a unified data discovery and comprehension portal:** The data asset catalog developed based on the unified semantic layer serves as the "data map" and "dynamic data dictionary" of the enterprise. Users can search, browse, and understand data assets from a familiar business perspective, such as by business theme, business process, or metric name. The catalog should not only display basic data information but also clearly reveal critical details such as the business context, calculation logic, data source, update frequency, quality status, data owner, and access permissions. This helps users quickly locate the required data and accurately understand its significance.



#### Semantic modeling empowers intelligent analysis in the era of LLMs: solving the problems of semantic scarcity and "hallucination"

**Provide precise and consistent "enterprise knowledge context" for LLMs:** The unified semantic layer defines and organizes the business rules, metric systems, dimensional hierarchies, and logical relationships between data in a structured and standardized manner. This acts as a "ground truth" and precise context environment for LLMs within the enterprise's business domain. When users submit queries, LLMs can reference the definitions in the semantic layer to disambiguate intent, understand real meaning, and generate responses that align more closely with business reality.

**Joining natural language and structured queries (Text-to-SQL translator):** For analysis tasks involving structured data, the unified semantic layer serves as a crucial "translator". It helps LLMs accurately map users' colloquial, natural language queries to the metrics, dimensions, and business objects defined in the semantic layer, converting them into executable backend data query prompts (such as SQL). This significantly improves the accuracy and robustness of text-to-SQL conversion, enabling non-technical users to interact with data through chat interfaces.



#### Materialized tables and adaptive acceleration based on the NoETL concept: improves development efficiency.



#### Rich ecosystem integration: unifies data exports and lowers data usage barriers.

# Unified stream-batch-incremental data processing by Stream Compute Service

**Challenges of enterprises** Enterprises face three major challenges in stream data development and operations: low development efficiency, high Ops costs, and inability to invoke AI models in real time. Writing complex Flink/SQL code relies heavily on the manual experience, making debugging and optimization time-consuming and labor-intensive. As data tasks scale, traditional Ops methods struggle to accurately identify anomalies, resulting in high false-positive rates and inefficient log analysis. Additionally, traditional AI analysis requires offline data export, leading to high response delays in risk control and recommendation scenarios. This prevents real-time model service from being directly applied upon data generation, hindering real-time workload value realization.

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**Solution** Stream Compute Service systematically addresses enterprise challenges through AI-driven code development and intelligent Ops, and deep integration of stream processing with real-time AI.

On the code development side, AI leverages Flink syntax and context semantics to auto-complete stream data processing code (such as generating window function logic) and provides optimization suggestions for SQL execution plans and resource consumption (such as index optimization and dynamic parallelism adjustment) to improve development efficiency. AI also detects syntax errors (such as operator configuration errors) and performance bottlenecks (such as memory overflow risks) in real time to reduce debugging time.

On the Ops and maintenance side, AI analyzes historical data and real-time metrics (such as throughput and backpressure) to predict anomalies (such as data heap) and issue early warnings. It optimizes resource assignment using task health scoring mechanism. AI also filters irrelevant logs to pinpoint root causes (such as service call timeout linkage) and automatically generates Ops reports to shorten failure location time.

Meanwhile, Flink CDC 3.3 natively supports dynamic AI model calls (such as OpenAI) during data stream processing to perform tasks like intelligent sorting, semantic analysis, or fraud detection on database change data. Combined with Retrieval-Augmented Generation (RAG) technology, it enables end-to-end low-delay processing, such as generating personalized recommendations the moment user behavior data is produced, breaking the efficiency barriers of offline analysis.

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**Use cases** A bank uses Flink CDC to capture transaction flows and dynamically calls AI models to analyze features like transaction amounts and IP addresses, intercepting high-risk transactions in real time. This reduces risk response time from minutes to seconds, increases daily interception rates, and lowers false-positive rates.

An e-commerce platform leverages RAG to generate dynamic recommendation lists based on real-time user browsing behavior, pushing them to the frontend, which boosts conversion rates and user engagement.

An internet company uses AI log analysis to automatically identify Flink task memory configuration issues, reducing task interruption rates and Ops manpower costs.

## Strengths



### Full-stack stream processing and AI capabilities

Realizes the platform with native deep integration of Flink CDC and AI models, supporting real-time AI service calls without data landing, ensuring ultra-low end-to-end delay.



### Validated by Tencent's industry-scale scenarios

Proven in Tencent's high-concurrency scenarios, handling millions of data processing tasks per second.



### Rich Tencent Cloud AI ecosystem

Integrates with mainstream model libraries like Tencent Cloud TI Platform and OpenAI, supporting rapid custom models for risk control, NLP, CV, and more.



### Intelligent full-lifecycle management

Offers closed-loop AI-assisted tools from code development (AI auto-completion) to task Ops (health scoring) and real-time decision-making (RAG enhancement), reducing technical barriers by over 50%.

## Data Lake Compute (DLC)

### Challenges of enterprises

Historically, big data and AI computing resources were siloed, forcing enterprises to purchase separate resources for each, which could not be shared. Data movement between Data and AI systems further reduced efficiency and increased costs.

### Solution

Tencent Cloud DLC integrates with the WeData platform to deliver a unified DataOps+MLOps solution. Building on mature data engineering linkages, it extends enterprise-grade machine learning capabilities, enabling data and computing resources for both data engineering and ML training/inference. This next-generation data platform offers an integrated, user-friendly, and cost-efficient solution, unifying data engineering, SQL, and ML features in a single platform. Key benefits include:



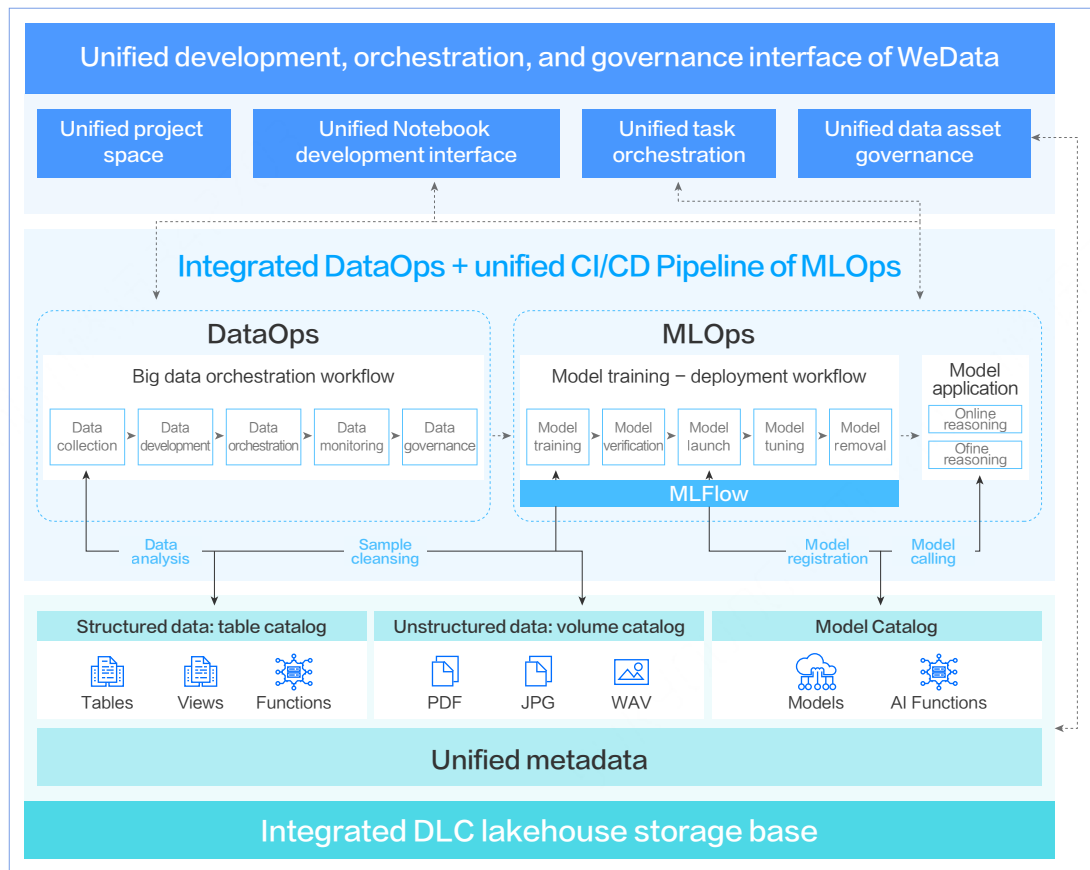
### Enhanced experience

A unified platform for both Data and AI application development.



### Lower costs

Shared computing resources between Data and AI reduce expenses.



## Use cases

In scenarios such as marketing automation, a large retail enterprise in Southeast Asian typically needs to not only generate wide-table data such as orders and user behavior based on traditional data analysis but also integrate data with ML training to generate ML applications such as sales forecasting and real-time recommendations. The enterprise has implemented an integrated DataOps + MLOps platform based on DLC. Tencent Cloud provides integrated Data + AI computing and training services to help users efficiently and cost-effectively train various ML models based on the same set of data and computing resources under traditional data engineering. Compared with the enterprise's previous deployment on a major overseas public cloud data platform, this solution achieved over 70% improvement in cost-performance efficiency.

### Unified CPU+GPU resource scheduling application

A Chinese new energy vehicle (NEV) manufacturer experienced a surge in vehicle sales and an increasing need for intelligent driving algorithm training and deployment. To meet these demands, the company sought to unify traditional data engineering and intelligent driving model training within a single platform. In addition, they aimed to enable unified scheduling of CPU and GPU resources across both data engineering and ML workloads to maximize overall resource utilization. By adopting the integrated Data + AI resource scheduling solution of Tencent Cloud based on its data lake architecture, the enterprise quickly implemented unified CPU+GPU scheduling. This approach effectively addressed the challenge of low resource utilization amid growing data volumes. Compared with maintaining separate scheduling systems for data and ML platforms, the unified solution improved resource utilization by 30%.

## Strengths

Provides a fully integrated, user-friendly, and cost-effective DataOps+MLOps solution. Compared to traditional data lake solutions from providers outside the Chinese mainland, DLC offers over 70% cost-performance advantages.



The unified execution engine seamlessly supports both data engineering and ML workloads, and common ML frameworks like Scikit-Learn, PyTorch, and TensorFlow.

The self-developed high-performance Meson engine doubles the performance of open-source Spark.

Highly flexible resource group management dynamically assigns computing quotas for data engineering, interactive analysis, and ML tasks.

## AI Data Lake Service (TCLake)

### Challenges of enterprises

Traditional data lakes store massive structured and unstructured raw data for Data + AI applications, but fall short in fragmented architectures across offline, real-time, and AI scenarios, lacking a unified framework for big data or AI engines calls. Key challenges include:



#### Dispersed metadata

Metadata for data lakes, data warehouses, and AI data sets/models is distributed across various platforms, leading to high costs for data version, lineage tracking, and license management.



#### Complex data management

Multiple system storage and workflow are used across offline, real-time, analysis, and AI scenarios. Diverse formats, small files, and expired and redundant data significantly increase governance complexity and governance overhead.



#### Difficult architecture optimization

Requires maintaining and optimizing multiple technical architectures. Each requires full-linkage management and tuning for storage systems, files, tables, and data, and struggles to keep pace with the flexible demands of modern AI businesses.

### Solution

Tencent Cloud's TCLake is an open, intelligent, and converged AI-native data lake infrastructure purpose-built for the Data + AI. It supports multimodal data storage and management, covering structured data, unstructured data, and AI models. It includes built-in unified metadata, unified batch and stream processing open table format, intelligent data optimization and load balancing, seamless integration with Tencent Cloud, and mainstream open-source Data + AI ecosystems. This enables enterprises to build a low-cost and unified data lake architecture tailored in AI era.

### Use cases

#### Lakehouse construction with unified batch and stream processing

The Lakehouse architecture merges the advantages of data lakes with the performance of data warehouses, offering a cost-effective, flexible, and diverse storage foundation. Users can deploy multi-computing engine applications (such as batch processing based on Spark, high-performance real-time data warehouse based on TCHouse, and ML applications based on SparkML) on the same lake-stored data. TCLake provides a unified metadata layer to consolidate Lakehouse assets, supports open table formats for unified batch and stream processing for data response needs across scenarios, and enables integration with multiple upstream engines. Its intelligent optimization and acceleration services dramatically reduce data maintenance costs and improve overall efficiency.

### Big data and ML integration

With TCLake's multimodal data management capabilities, enterprises can rapidly build unified big data and ML applications. Preprocessed training data from upstream engines like Spark can be registered in the metadata catalog. Downstream ML frameworks such as PyTorch and TensorFlow can directly read and train on this data. Trained models can be registered back into TCLake for unified lifecycle management. It also offers accelerated services for varied Data + AI workloads, greatly improving efficiency in AI application development and management.

## Strengths



### Multimodal unified metadata

Manages structured data in the big data domain, unstructured sources in the AI domain, and model assets via a unified metadata model and API, supporting diverse structured data sources and lifecycle management for ML model files.



### Open table format for unified batch and stream processing

Built-in TC-Iceberg table format, an extension based on Apache Iceberg with non-intrusive enhancements, supporting second-level data responsiveness and multiple open formats.



### Open engine ecosystem

Compatible with Tencent Cloud big data services and mainstream open-source computing engines such as Spark, Flink, Ray, and Pandas. It ensures equal access to structured and unstructured data in the catalog.



### Intelligent data optimization

Provides fully managed services for lifecycle management, data compaction, and data reorganization (clustering and Z-order). Users are freed from underlying data maintenance and optimization.



### Multi-load performance optimization

Provides advanced features like operator conversion and hot data acceleration to meet varying Data + AI business response demands across scenarios.

## Tencent Cloud Elasticsearch Service (ES)

### Challenges of enterprises

Traditional keyword-based search is limited to text data and fails to support images or videos. More importantly, it lacks semantic understanding, treating terms like computer and PC as completely different concepts. Over-reliance on LLMs may produce misleading hallucinations due to outdated training data, exhibit inefficiencies in domain-specific knowledge processing, lack deep domain insights, and demonstrate limited reasoning capabilities. Against this background, Retrieval-Augmented Generation (RAG) has emerged and become a major trend in the era of LLMs.

## Solution

RAG significantly enhances the accuracy and relevance of content by first retrieving relevant information from a vast database of specialized documents before the LLM generates responses, and then using the specialized information to guide the results generated by the LLM. Enterprises enhance their data AI readiness by using RAG as a Service (RAGaaS), and seamlessly integrate technologies such as vector search and graph databases to shorten development cycles and improve the quality of applications. Adaptive governance models based on active metadata can be used to balance centralized and distributed management and ensure the accuracy and compliance of AI outputs. For example, real-time metadata monitoring can be performed to prevent malicious use and privacy breaches.

ES is the world's No.1 vector search engine, ranked by DB-Engines. Vectorization search encodes the features of text, images, or videos into high-dimensional vectors. By calculating distances (such as cosine similarity, dot product, Euclidean distance) in vector space, it determines content similarity, vastly expanding search scope into multi-modal domains. When integrated with LLMs, ES can power RAG applications based on knowledge bases.

## Use cases

### Vector search

Examples include map search in Game for Peace, product recommendation on an e-commerce platform, and image moderation on a social media app, all powered by ES's multimodal and semantic search capabilities.

### RAG applications

Tencent products such as WeRead Reading (AI Q&A), ima (intelligent workspace), and Tencent Meeting Assistant leverage hybrid text and vector search for enhanced RAG applications and support billions of vectors with millisecond-level response.

## Strengths

Vector search excels at handling ambiguity, capturing semantics, and supporting cross-text, cross-language, and cross-modal queries. Traditional text search offers exact matching for short text, low computational cost, and high interpretability. Hybrid search combines both strengths:



### Higher accuracy

Concurrent keyword and vector queries improve result relevance and trustworthiness.



### Greater diversity

Utilizes the semantic richness of vector search to return a wide variety of results, offering users more options and addressing diverse search intents.



### Enhanced capabilities

Keyword-based logical operators, sorting, and filtering capabilities enable complex query scenarios, such as multi-condition, multi-attribute, and multi-collation searches, thus greatly expanding query capabilities.



### Better interpretability

Uses keyword-based text highlighting and relevance scoring to deliver explainable results, showing exact match positions, matching content, and query-document relevance, enhancing user trust and understanding.



#### At the kernel level

hybrid search has conducted deep optimizations tailored for typical vector scenarios, such as shard architecture optimization, parallel query execution, and Lucene query cache lock improvements. These enhancements ensure sub-millisecond average response delay for billion-scale vector indexes, delivering a 3x to 10x performance boost for hybrid search. Self-developed by Tencent Cloud ES, v-pack plugins support dynamic pruning of redundant vectors in row-based storage, achieving over 70% lossless compression of vector storage. Tencent Cloud ES pioneers the use of int8\_only\_hnsw index, compressing high-precision vectors to int8 format, saving 90% storage. Regarding the re-ranking strategies, it supports adjustable-weight RRF, normalized Score, and GPU-accelerated Rerank model service calls.

## TBDS multimodal data lakehouse

### Challenges of enterprises



#### Explosion of multimodal data

By 2024, enterprise data in China is growing at a rate of 25% per year, with over 70% being unstructured. This includes more than 10 data modalities, such as IoT device data (industrial sensors), audio and video streams (live-streaming e-commerce), biometric data (medical imaging), and spatial data (HD maps for autonomous driving).



#### Demand for deep multimodal analysis

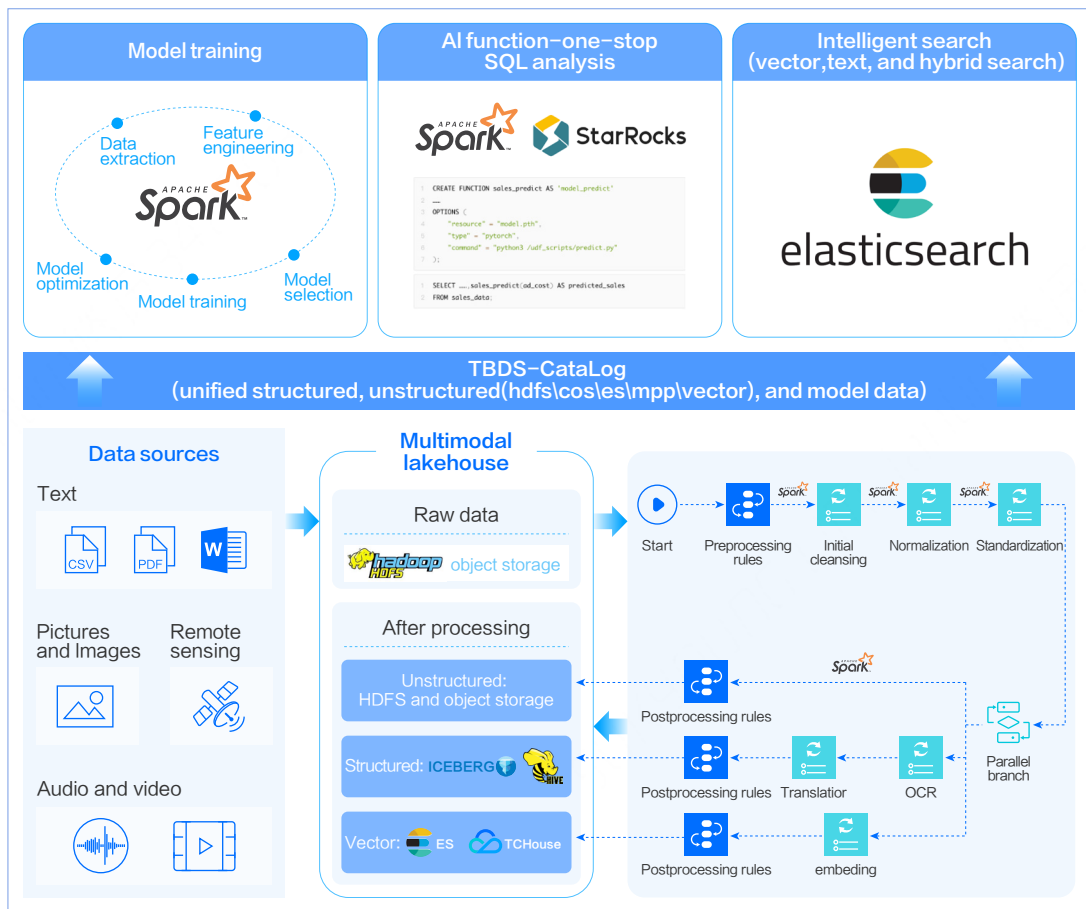
Research based on delivered enterprise shows that over 60% of enterprises have made cross-modal analysis a top priority for digital transformation in the era of LLMs. Enterprises are actively exploring how to more efficiently access, store, manage, and apply multimodal data to unleash greater value from unstructured sources. Leading-edge enterprises are also investigating multimodal data alignment and cross-domain feature extraction.



#### Building enterprise private knowledge bases

As challenges like GenAI hallucinations and compliance issues emerge, enterprises require more rigorous solutions when adopting AI technologies. They need to convert diverse data modalities into a knowledge base while ensuring data privacy and controlled usage compliance.

## Solution



### Multi-modal storage engine

TBDS supports unified access and raw data storage for 20+ data modalities (including various text, images, audio/video streams, and spatial data) through HDFS distributed cloud file storage and object storage. During data cleansing and processing, structured data can be uniformly stored via services like Iceberg and Hive. Processed vector data can be efficiently stored and searched using products like ES and TCHouse.



### TBDS-Catalog

Through unified metadata indexing technology, TBDS-Catalog enables metadata registration and management of cross-storage media (such as object storage, databases, and data lakes). It provides unified descriptions for 20+ modalities, spanning text, images, videos, time-series data, as well as structured and semi-structured data from traditional data lakes, data warehouses, and message queues. It also accommodates vector data and model data in the GenAI era. This ensures unified metadata governance for data across multi-cloud, cross-region, and multi-organization scenarios.



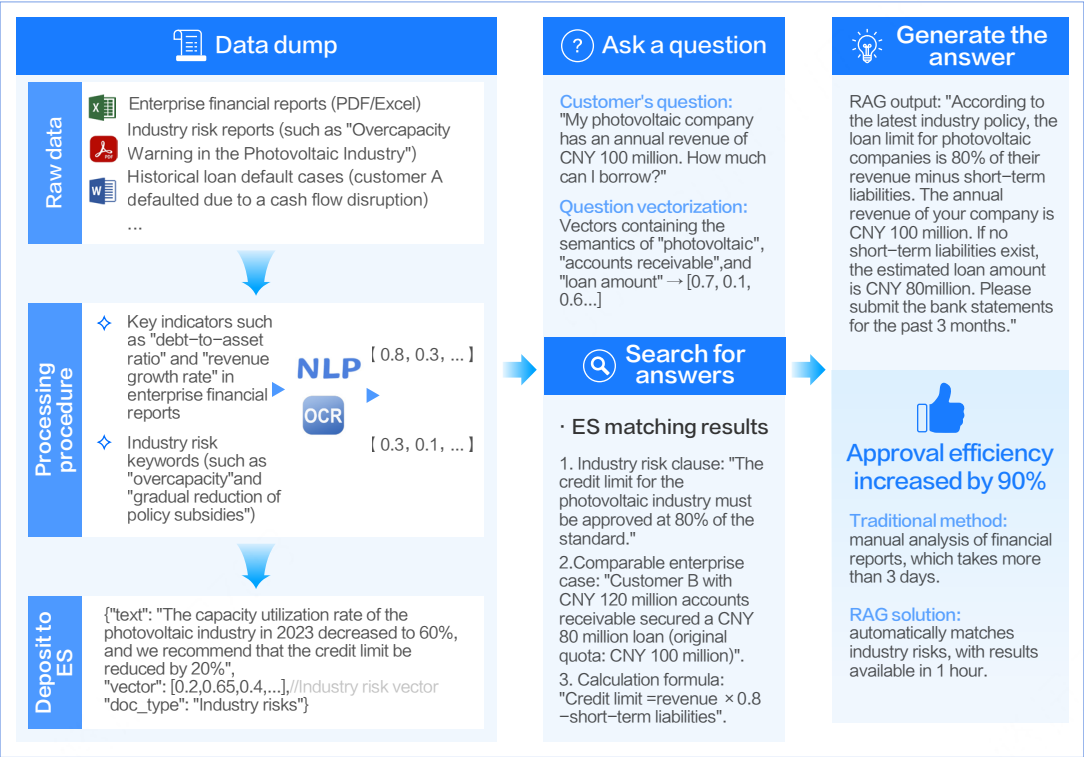
### Cross-modal analysis application

TBDS empowers upper-layer applications via three core capabilities: model training, AI functions, and intelligent search. In the model training scenario, TBDS-Catalog enables unified data access and version control for training data scattered across different storage media, dramatically increasing training efficiency. In the AI function scenario, trained models can be encapsulated as SQL functions and directly called during SQL analysis, eliminating coding requirements and enabling direct usage by data analysts and BI engineers. Intelligent search supports hybrid search across structured data, text data, and vector data with capabilities like semantic similarity search, text-to-image search, and image-to-image search.

Use cases

In the wave of digital transformation across industries, multimodal data has become a core strategic asset driving innovation. Whether it's policy documents and financial reports, audio–visual interactions in finance, medical imaging and genomic data in healthcare, classroom videos in education, or live content and consumer behavioral tracks in retail, enterprises are increasingly focused on unlocking the value of these diverse data types.

For example, in banking scenarios, multimodal data management and fusion analysis are now widely applied across risk control and compliance, knowledge Q&A, and product management. These scenarios typically involve numerous text documents, such as enterprise financial reports (PDF), industry development reports (PDF), industry risk reports (PDF), banking regulatory policies, user data privacy policies, and historical loan default cases (Word). Machine–based models' processing of these documents far outperforms manual handling in efficiency. Take loan approval as an example:



Firstly, enterprises gather and clean documents such as financial reports, industry analysis reports, and historical loan default cases. Then, NLP/OCR technologies are used to extract structured information. Extracted content is vectorized via LLM call capabilities and stored in Elasticsearch. Secondly, customer queries are vectorized, with relevant materials via vector similarity search. Thirdly, retrieved documents are fed into LLMs, which synthesizes an answer based on the model's generative capabilities. Compared to manual analysis, this RAG–based automatic approval pipeline improves overall approval efficiency by more than 90%.

Strengths

**Hybrid vectorization lakehouse storage engine**

Built on Apache Iceberg, this engine forms a unified foundation for multimodal data, supporting the strong consistency of structured/semi-structured data through ACID transactions and full-linkage versioning. Tencent Cloud's self-developed vectorization storage engine for Iceberg addresses performance bottlenecks in high-dimensional feature search for unstructured data. It enables federated SQL queries across structured logs, semi-structured metadata, and unstructured vector data and breaks down traditional barriers between data lakes, object storage, and Tencent Cloud VectorDB.



#### Unified semantic multimodal metadata

TBDS links metadata across Tables, Files, Models, and Functions, enabling unified semantic mapping between data assets and computing resources. This eliminates metadata silos common in traditional structure that treats structured data, unstructured files, ML models, and functions as isolated entities. It empowers upper-layer applications with cross-modal federated metadata services and unified data asset contents.



#### Intelligent lakehouse platform Ops agent

Built on the "Perception – Decision – Execution – Evolution" intelligent closed-loop paradigm, the TBDS intelligent agent is driven by self-developed Ops LLMs and Ops knowledge base for big data operations. It manages the entire lifecycle of system faults, using a multi-agent architecture that supports automated operations from prediction and alerting to self-healing and resolution. Continuous learning from historical incidents enables iterative capability enhancement.

## TCInsight

### Challenges of enterprises

As digital transformation accelerates, the complexity of enterprise IT infrastructure has increased at an exponential rate. Traditional Ops approaches are struggling to manage the rising complexity of platforms, accompanied by soaring Ops costs. Intelligent autonomy of big data platforms based on AIOps is becoming a pivotal approach to enhance Ops efficiency, reduce costs, and ensure business continuity. The intelligent autonomous system for data platforms integrates Ops data such as logs and metrics, and leverages traditional machine learning combined with AI Agent to implement features such as fault prediction, root cause analysis, and automated remediation. The ultimate goal is to improve the Ops response speed from an hourly level to a millisecond level, reduce manual intervention to below 10%, and usher in an era of "autonomous driving" for big data Ops.

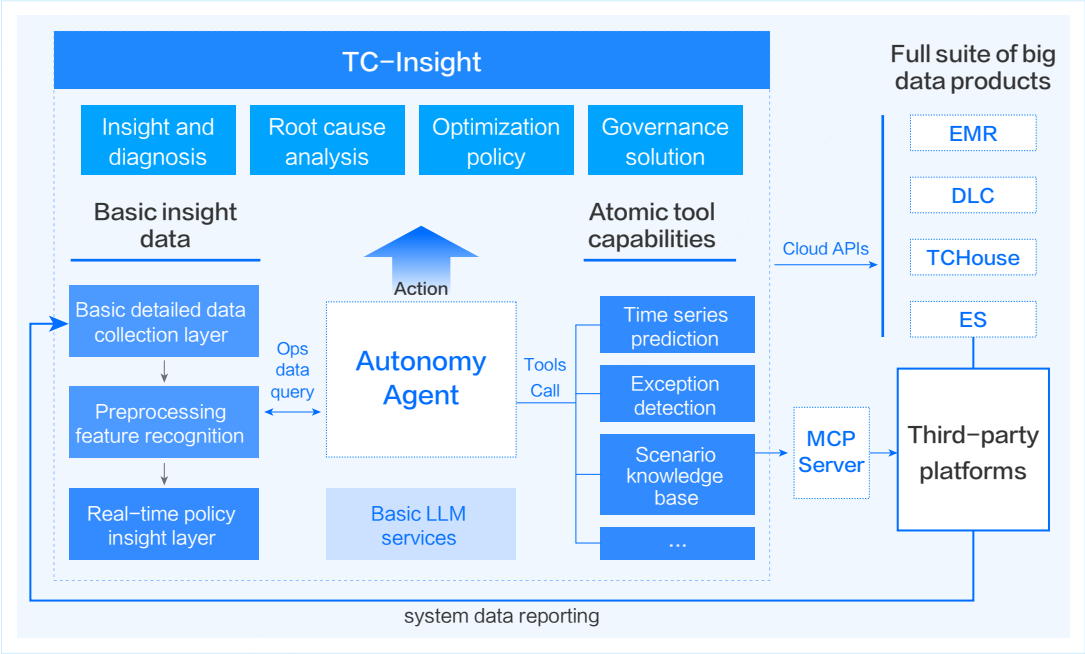
Traditional Ops face significant challenges, including over-reliance on expert experience for failure location, low efficiency in exception diagnostics, and lack of dynamic awareness in resource scheduling. These issues lead to prolonged system recovery time and severe resource waste.

Root cause analysis and issue troubleshooting heavily rely on the experience of Ops engineers, while querying anomalies or performance tuning requires expert-level data developers. In traditional Ops, 70% of system faults require troubleshooting across more than three infrastructure layers, with an average Mean Time to Recovery (MTTR) of over 4.5 hours, relying on inefficient manual exception diagnostics. In addition, traditional alarm-based Ops systems suffer from persistently high false alarm rates, typically ranging from 30% to 50%. For example, a logistics enterprise receives over 500 alarms per minute, of which 99% are invalid. AIOps, through intelligent convergence, increases the proportion of valid alarms to 90%. Finally, traditional big data Ops systems lack dynamic awareness of resource scheduling, with 40% of enterprises experiencing server resource waste or overload issues. The balance between Ops costs and business agility is difficult to achieve.

The intelligent autonomous system of data platforms can deliver significant operational efficiency improvements for industry-leading enterprises with advanced digitalization, complex IT systems, and stringent business continuity requirements. For example, several leading short-video platforms in China handle over 100 million daily active users, with their big data platforms generating over 100 TB of log data daily. The annual Ops costs account for over 30% of their total technology expenditure. With the use of intelligent autonomous systems, Ops costs can be reduced by 90%. In addition, through intelligent task optimization, the utilization of computing resources can be improved by over 20%.

**Solution**

Tencent Cloud TCInsight is an intelligent autonomous solution designed for the complete suite of public cloud big data products. Based on traditional ML models and AI Agent, it aims to deliver automated system detection, analysis, and optimization capabilities without human intervention, thereby achieving the "Autonomous Driving" vision for big data Ops. The core value of TCInsight lies in simplifying complex big data management tasks by using intelligent technologies. Traditional manual Ops are inefficient and costly, whereas TCInsight leverages AI to achieve system auto-optimization, cost analysis and tuning insights, and 24/7 autonomous Ops and governance, and health diagnostics. By deploying AI agents instead of human troubleshooting, TCInsight ensures faster and more accurate problem discovery and resolution, directly addressing key enterprise pain points.



Typical application scenarios of Tencent Cloud TCInsight include:

-  **autonomous optimization**  
optimizes computing jobs, data storage lifecycle, data distribution, and resource scheduling.
-  **Automated FinOps**  
optimizes resource procurement strategies based on actual usage patterns to achieve superior cost efficiency.
-  **Automated system Ops**  
System fault analysis and troubleshooting rely heavily on the experience of Ops engineers and often take more than 4.5 hours. Customers generally hope for AIOps to replace human labor.
-  **Proactive awareness and handling of Ops issues**  
Predictive intelligence is used to identify potential system Ops and task issues in advance and automated processing is used to enhance the overall system efficiency, such as detecting parameter configuration issues in periodic ETL jobs and automatically fine-tuning them before the next execution.



## Use cases

### Computing and SQL tuning

Each month, TCInsight helped a customer identify more than 50 inefficient queries, such as large table scans with 100K partitions, full partition scans, and abnormal resource usage. Based on tuning insights, operations were automated via API integration with user development platform, resulting in significant improvements in task stability and resource efficiency.

### HDFS storage governance

For users using HDFS local storage, TCInsight identified small file partitions with less than 10 MB through daily storage analysis. With automated recommendations, the user optimized storage capacity from 4.81 PB to 3.32 PB and decommissioned 17 resource nodes, reducing costs.

### Intelligent autonomy of third-party data platforms

MCP Server provided by TCInsight can be used to implement rapid autonomous takeover of third-party data platforms. Various intelligent autonomy features of TCInsight can be integrated into third-party data platforms, enabling traditional platforms that lack big data autonomy capabilities to quickly upgrade to AIOps.

## Strengths

Tencent Cloud TCInsight delivers more intelligent auto-tuning and cost optimization powered by AI Agent. Compared with similar products from cloud service providers in China, it provides unified multi-product support. Compared with third-party SaaS service vendors, it offers the advantage of integrated deployment. It provides the following core competitive advantages:



### More precise

TCInsight is combined with AI Agent to achieve significant improvements in both optimization accuracy and coverage rate. For example, TCInsight resolves 80% of performance-critical cases, such as Spark job tuning, HiveSQL optimization, and query parameter adjustments, that traditionally caused failures or delays.



### More comprehensive

TCInsight provides coverage capabilities for multiple computing engines (Hive, Spark, and Impala), storage engines (Impala tables and SR tables), and products (TCHouse and ES).



### More open

TCInsight can provide independent intelligent autonomy capabilities and can also be integrated into customers' own data platforms and Ops systems.

## Tencent CloudDB AI service

### Challenges of enterprises

With the deepening application of AI in IT Ops, enterprises face multiple challenges in database management: DBA talent shortage intensifies Ops pressures, while manual operations lead to misjudgments and inefficiencies. Lack of end-to-end control from SQL development to change deployment and production monitoring, particularly in advanced scenarios like code-level risk prediction and precise fault tracing. There is an urgent need for an enterprise-grade database AI service platform to address these Ops challenges.

## Solution

Tencent CloudDB AI Service Platform focuses on intelligent database Ops for enterprise users. Centered around the two core scenarios "Run SQL Well" and "Write SQL Well", the platform is deeply integrated with Tencent Cloud's database PaaS/SaaS capabilities, delivering scenario-based, end-to-end intelligent agent services, enabling autonomous data value realization.

The product builds a super DBA agent system through autonomous Ops and developer assistance, establishing a full-linkage SQL risk prevention framework covering SQL development, DDL changes, and production monitoring.



### Pre-execution SQL risk prediction

Associates with actual production code to assess risks before SQL deployment.



### DDL change assessment

Replays real traffic with audit logs to evaluate DDL change impacts.



### High-load monitoring

Directly traces slow SQL-induced database stress to specific code lines.

This intelligent system significantly reduces reliance on DBA manpower and dramatically improves troubleshooting speed and accuracy.

The hybrid deployment architecture supports elastic pay-as-you-go for public cloud users and enables data-localized closed loops for sensitive sectors (such as finance).

By injecting AI across the database management lifecycle, it pioneers an "Operations-as-a-Service" model, driving the upgrade from manual operations to fully autonomous database management.

## Use cases

### Background

A major customer faced significant challenges during online code deployment amid rapid business expansion. SQL statements introduced during code releases caused significant performance degradation of the full database, severely impacting business operations. The customer sought to preemptively identify risky SQL before deployment.

### Challenges

Developers lacked comprehensive database knowledge (such as index design, composite index failures). They relied heavily on ORM frameworks and had little visibility into the final executed SQL. Once faults occurred, traditional troubleshooting could not trace issues back to the source code.

### Solution and results

The customer deployed SQL risk forecasting Agents in a pilot workload. Using a dynamic SQL parser and a risk analysis engine, they achieved:

**Comprehensive coverage:** Scanned 220K code lines, extracted 1,500+ SQL statements (80% coverage), bypassing ORM black-box limitations.

**High-precision detection:** Flagged 200+ high-risk SQLs, including missing indexes, ill-ordered query conditions, with 92% accuracy in optimization suggestions validated by customers.

**Sustainable defense:** Maintains 80% code scan extraction and 92% accuracy across 4 subsequent releases, preventing 12 potential risks with zero database performance incidents.

This solution not only bridges the database knowledge gap for developers but also shifts risk control upstream to the development phase, forming a closed loop of "code-scan-fix" and delivering quantifiable gains in business stability.

## Strengths



### Full-linkage SQL risk control system

Covers the full lifecycle from SQL development to DDL changes and production monitoring.



### Result-oriented intelligent agents

- Pre-deployment SQL risk prediction
- Code-line level traceability
- End-to-end outputs
- Autonomous 24/7 operation



### Secure and controllable architecture

Modular design supports private deployment, enabling customer-side control over data, models, and code.



### Scenario-based capability integration

Deep integration with Tencent CloudDB PaaS/SaaS product stack (such as CDB, TDSQL, DBBrain), offering end-to-end scenario-based services.



### Hybrid deployment support

Flexible public/private cloud delivery.



### Data visualization

Multidimensional agent usage analysis, including cost, call volume, tenant distribution, and trend forecasting.

## Tencent Cloud VectorDB

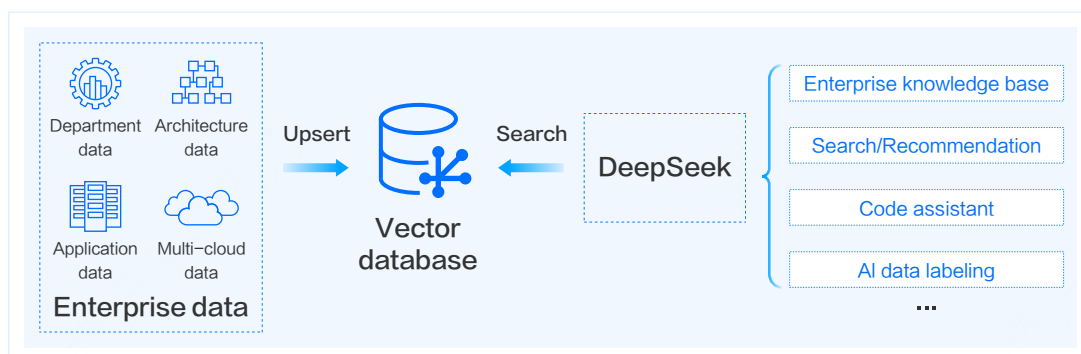
### Challenges of enterprises

DeepSeek is spearheading a new wave in the AI technological revolution. As enterprises actively embrace this transformation, the internal enterprise data formats and search demands are undergoing a fundamental shift. With the explosive growth of LLMs and GenAI, traditional keyword-based search methods can no longer meet the evolving requirements for semantic understanding and multimodal search. At the same time, enterprise data is growing exponentially, with both structured and unstructured data coexisting. This creates an urgent need for an intelligent search solution capable of processing heterogeneous data from multiple sources in a unified manner.

### Solution

The key to successful enterprise implementation of Data + AI lies in deep integration between enterprise data and LLMs, making Tencent Cloud VectorDB the essential solution for LLMs to access enterprise data. The combination of LLMs + Tencent Cloud VectorDB has become the perfect partnership for deploying AI applications in enterprises.

Convert structured and unstructured data of enterprises into vectors. Store these vectors in Tencent Cloud VectorDB to establish an internal data hub. This implementation overcomes traditional search limitations and connects multi-source heterogeneous data with AI applications. This approach represents the best practice for advancing next-generation retrieval technologies, enabling comprehensive enhancements to application systems such as enterprise search and intelligent recommendations. By integrating LLMs, organizations achieve measurable efficiency gains and improve user experiences.



Tencent Cloud VectorDB ranks as the first enterprise-grade, self-developed distributed DBMS certified by China Academy of Information and Communications Technology. It builds upon Tencent Group's extensive technical expertise, operating reliably in over 60 core workloads within Tencent (including Tencent Video, Tencent Meeting, QQ Music, and other popular applications). Supporting over 850 billion vector retrieval requests per day, it can store hundreds of billions of vectors, five million queries per second (QPS), and millisecond-level query latency. As a "data hub" in the AGI era, VectorDB specializes in storing and retrieving unstructured data such as documents, images, audios, and videos in retrieval/recommendation and AIGC scenarios. VectorDB acts as the ideal partner for LLMs to deploy AI applications.

## Use cases

### Background

A leading education customer prioritizes their AI Learning device development, aiming to utilize the RAG development framework to integrate accumulated textbook data and teaching aid resources with AI. This project creates an AI assistant for learning devices. This approach leverages the value of massive course and exam resources to provide immediate precise student answers. The solution enhances online teaching efficiency while reducing labor expenditure.

### Challenges

In practical application processes, clients intended to employ traditional keyword search schemes due to LLMs' hallucination issues. However, students' questions are typically broad and non-standardized, resulting in only 60% accuracy during effectiveness tests. Through preliminary research and testing, customers adopted Tencent Cloud VectorDB as a supplemental knowledge base for LLM. The retrieval system centers on semantic-level retrieval powered by vector data.

### Solution and results

1. Following implementation, customer validation confirms that vector retrieval alone addresses 95% of scenarios, outperforming keyword-based search schemes.
2. Resource-intensive embedding model selection and cluster maintenance led customers to adopt Tencent Cloud VectorDB's native embedding feature. This improves workload access efficiency by 80% and reduces client costs by 90%, and has maintained zero stability issues since deployment.
3. Leveraging the AI suite functionality of Tencent Cloud VectorDB, it offers a comprehensive documentation retrieval solution that allows users to upload unprocessed files, enabling the construction of private knowledge bases in hours, a reduction from months, while eliminating lengthy setup processes and significantly enhancing knowledge onboarding efficiency.

## Strengths



### Decades of internal technical expertise

Utilizing the core OLAMA vector retrieval engine, it processes over 850 billion daily vector queries across Tencent's 60+ core services.



#### "First" Authoritative Certification

The first enterprise-grade self-developed Distributed DBMS in China to be certified by the China Academy of Information and Communications Technology.



#### Large-scale, high-performance capabilities

A single index can accommodate data scales of up to hundreds of billions, achieve five million QPS, and provide millisecond-level query latency.



#### High availability and reliability

Distributed clusters, multi-availability zone (AZ) deployment, and multi-replica ensure enterprise-grade availability and reliability.



#### Embedding and AI suite features

supports rapid vectorization and automatic storage for a variety of unprocessed text and documentation files.



#### Hybrid retrieval enhances recall rates

Capable of semantic and keyword-based bidirectional retrieval. In specific scenarios, hybrid retrieval achieves a recall rate of 95%, a 30% improvement compared to traditional keyword search approaches.

## TDSQL-C

### Challenges of enterprises

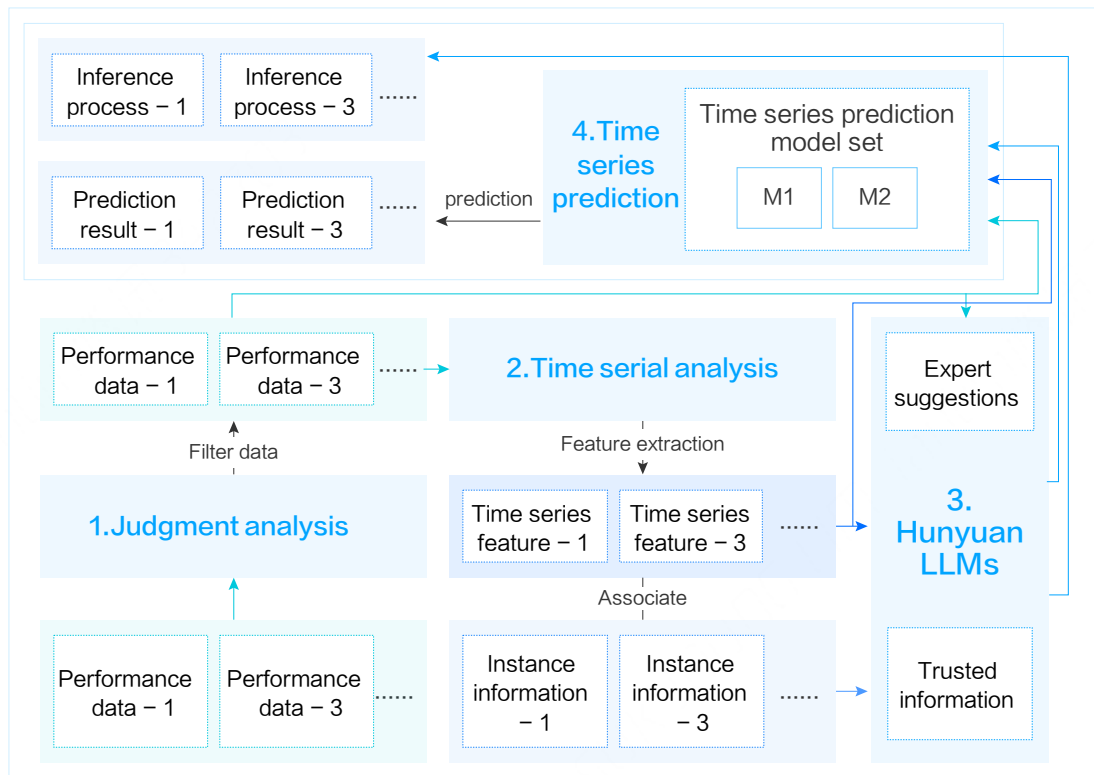
Handling substantial peak data traffic often leads to challenges such as resource insufficiency and reduced efficiency due to delayed scale-in and scale-out capabilities. Frequent scaling adjustments may provoke fluctuations, significantly increasing operational costs for enterprises. Additionally, unoptimized SQL statements (e.g., full table scans and redundant subqueries) lead to poor query performance, resulting in response delays under high-concurrency scenarios. The absence of automatic tuning tools or expertise limits effective use of overlays, indexing, and query rewriting methods, escalating database load.

### Solution



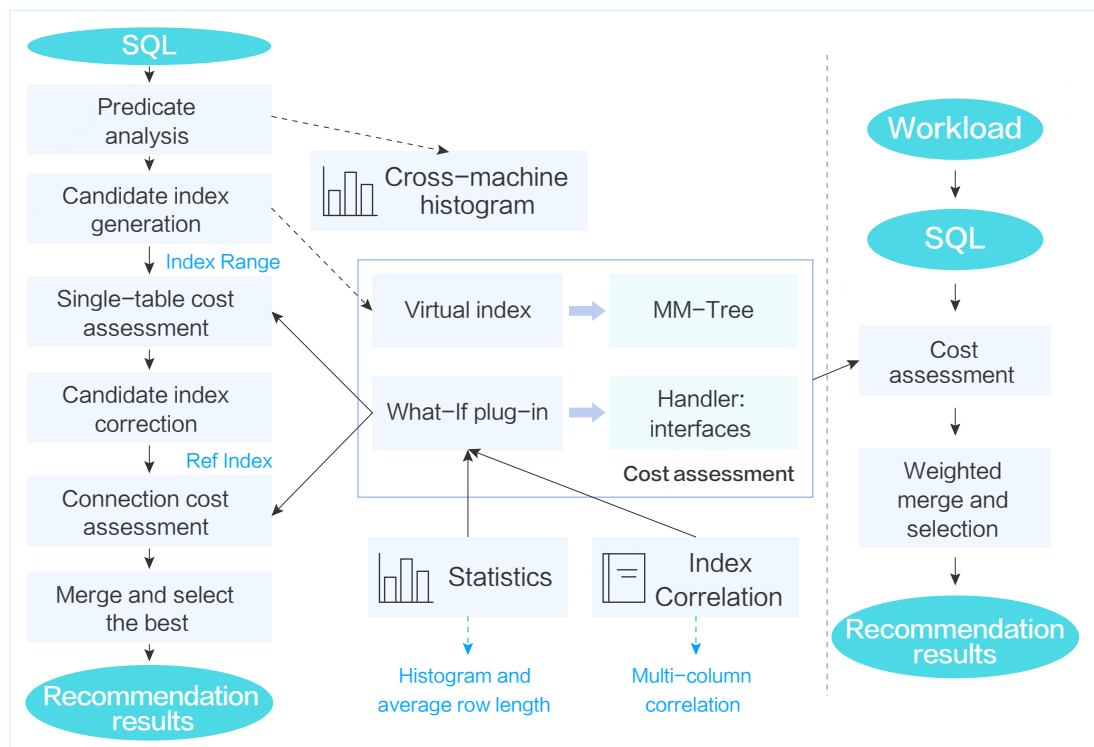
#### Predictive auto-scaling capability

Anticipate workload fluctuations in advance and customize the proactive auto-scaling timing. Eliminate the risk of scale-out delay attributed to reactive auto-scaling. Leveraging ML technology, the database forecasts future resource demands based on historical workload data and dynamically adjusts resource allocation. TDSQL-C employs a load prediction model grounded on time-series analysis to monitor CPU, memory, disk I/O, and other metrics in real time, optimizing auto-scaling policies through the integration of the trial-and-error mechanism inherent to Reinforcement Learning (RL). In AI workload scenarios, the system automatically balances GPU training and database memory utilization to enhance resource efficiency.



#### Smart index recommendation

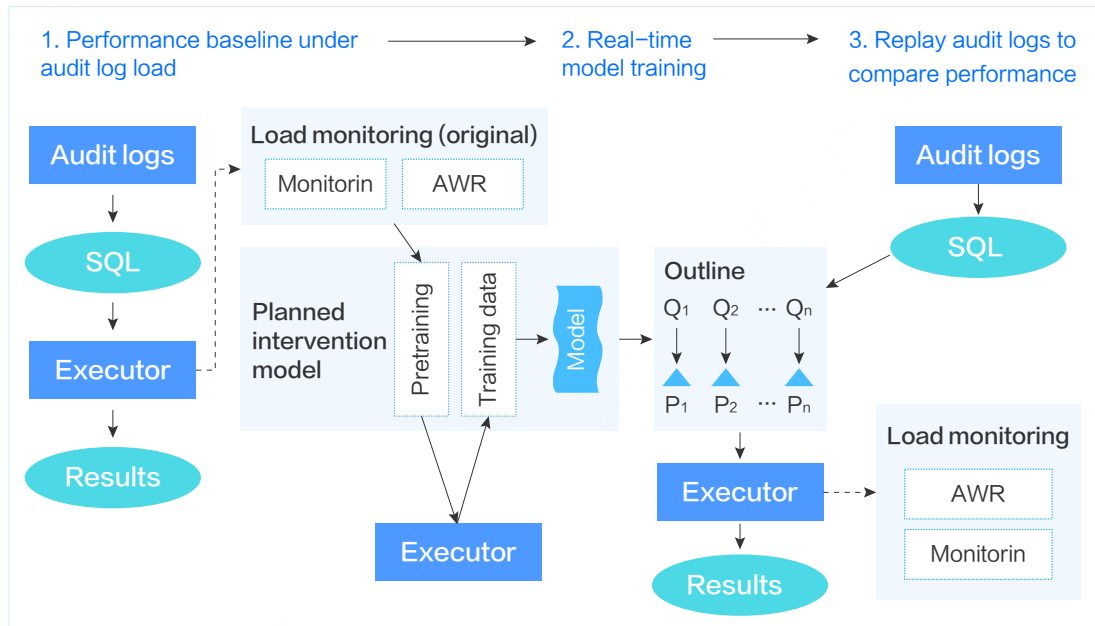
Establish single SQL index recommendation capabilities utilizing statistical information and develop full-linkage cost-based index recommendations through predicate reordering and cost estimation refinement. Generate virtual indexes dynamically and perform cost comparisons in real time to offer users the optimal index solution. Develop deep learning model to dynamically adjust index structures. Automatically select B-tree, bitmap, or columnar indexes based on the data access pattern, while utilizing index compression techniques to balance storage efficiency and query performance.





### AI-driven optimizer

Automatically tune query plans using feedback information, intelligently extrapolate more strategies, bind to the most effective execution plan, reduce average workload, enhance plan stability, and minimize operational costs. Support automatic parameter tuning for diverse SQL statements, encompassing semi-join execution strategies, predicate pushdown, and sort optimization. Expand optimization coverage from single-table to multi-table, scaling from local optimal approaches to global optimal strategies. The execution results serve as feedback within the learning process, identifying optimal plans with minimal exploration attempts. ML methods excel at capturing intricate features to determine the best plan using parameters and data.



## Use cases

### Predictive auto-scaling capability

Designed for high-load scenarios such as e-commerce sales or financial transaction peaks. For instance, e-commerce platforms leverage AI to predict peak attack traffic (e.g., 300 Gbit/s during the Double 11 Shopping Festival), dynamically allocating Anti-DDoS Advanced resources in conjunction with real-time workload to achieve second-level scaling and load balancing, thereby ensuring service stability throughout the promotional period.

### Smart index recommendation

Designed for complex query optimization and workload tuning scenarios. Evaluate the optimization effectiveness without physically creating indexes, assisting retail enterprises in enhancing their inventory query efficiency. Single-SQL query efficiency gains up to 400% through statistics-driven index recommendations.

### AI-driven optimizer

For intricate OLAP operations such as multi-table joins and aggregate queries, the AI-driven optimizer, utilizing reinforcement learning, produces optimal execution paths integrated with adaptive cardinality estimation, substantially enhancing data analysis efficiency. For scenarios like e-commerce flash sales and financial transactions, the AI-driven optimizer formulates optimal query plans in real time, significantly improving workload performance efficiency.

## Strengths

The world's inaugural predictive auto-scaling capability has been integrated into the serverless technology domain. Leveraging sub-second auto-scaling capabilities, it enables preemptive scaling before workload peaks, thereby fully automating Ops and maintenance. Within the realm of database autonomy, integration of AI-driven optimizer and adaptive indexing architecture comprehensively boosts SQL execution efficiency. Single-query performance improvements reach up to 400%, with average latency reduced by 42.2%.

# Tencent Cloud Agent Development Platform (TCADP)

## Challenges of enterprises

Enterprises currently encounter numerous data obstacles in the realms of knowledge management and AI applications:



### Knowledge fragmentation and inefficient management

Enterprise documentation often exhibits a complex structure (e.g., cross-page tables, mixed text and images, and multimodal content). Traditional parsing tools struggle to accurately extract unstructured data, leading to delays in knowledge base updates and low retrieval efficiency.



### Complex business scenarios

In fields such as customer service, finance, and education, there is a growing need to handle diverse queries. However, traditional AI systems rely on manual configuration, resulting in limited flexibility and scalability as well as high costs.



### Data silos and security risks

Multi-source data resides in isolated systems without unified governance. Besides, Sensitive information, such as financial data, is prone to leakage due to technical vulnerabilities.



### High AI adoption barriers

Implementing LLM requires specialized teams for data cleaning, model tuning, and process choreography, which creates technical and cost challenges for small and medium-sized enterprises.

## Solution

The Platform provides versatile application development frameworks, including LLM + RAG, Multi-agent, and Workflow, pre-configured with selected official plugins and MCP plugins. It provides an all-in-one toolchain for application configuration, testing, evaluation, and publishing, helping enterprises easily build large model applications.

1. The Platform includes built-in best practice workflows. By only importing documents or Q&A pairs, the LLM can integrate with enterprise knowledge to achieve a more stable and accurate Q&A results. This is ideal for precise Q&A scenarios, such as enterprise knowledge services and product consultations.

2. The Platform uses the designated workflows to handle all user conversations. For more personalized application execution process, workflows allow users to drag-and-drop various atomic components (such as LLM nodes, knowledge Q&A nodes, parameter extraction nodes, and tool nodes) to customize and orchestrate the business processes.



3.The Platform supports switching to agent mode, where the LLM autonomously breaks down complex tasks and plans paths based on prompts. It actively selects and utilizes tools, corrects and reflects errors, making the responses more flexible. Additionally, multiple agents can be added to collaborate on complex tasks.

## Use cases

### Case study 1: AI-driven customer service upgrade (First Automobile Works Toyota, FAW Toyota)

When deploying AI-driven customer service LLMs, enterprises encounter challenges such as disconnected proprietary knowledge, slow responses, vague answers, and high knowledge base data organization costs. FAW Toyota overcame these issues by using RAG technology from the Tencent Cloud Agent Development Platform, combined with OCR, multimodal, and long-text processing capabilities, enabling precise matching with their proprietary knowledge base and automatic extraction of historical customer service data. As a result, their independent customer service resolution rate increased from 37% to 84%. The upgraded AI-driven customer service now supports FAW Toyota's official website, app, mini program, WeChat official account, and other channels, handling an average of 17,000 consultations monthly, significantly improving service efficiency and user satisfaction.

### Case 2: Enhancing financial business efficiency (Chongqing Rural Commercial Bank)

Financial product knowledge is complex, and traditional financial services often suffer from low efficiency, imprecise risk control, and underutilized data. By leveraging Tencent Cloud Agent Development Platform's networking search, RAG, and knowledge base capabilities, Chongqing Rural Commercial Bank launched its AI assistant "AI Xiaoyu" based on the DeepSeek model embedded in Tencent Cloud's platform, which provides three primary solutions: 1) AI-driven risk management: dynamically identifying fraudulent activities and enhancing risk warning capabilities; 2) Scenario-Based Finance: Offering minute-level responses in AI-driven customer service and merging knowledge base data to provide personalized wealth management suggestions; 3) Data-driven decision-making: leveraging LLMs to mine internal financial data insights, optimizing credit assessments and market strategies. As a result, "AI Xiaoyu" has significantly boosted employee collaboration, facilitated the intelligent upgrading of financial services, and strengthened its brand value as a "Digital Rural Commercial Bank."

### Case 3: Pharmaceutical retail services (DaShenLin Pharmaceutical Group Co., Ltd.)

As a leading enterprise in the pharmaceutical retail industry, DaShenLin faced challenges such as low internal office collaboration efficiency, slow responses in domain-specific business scenarios, and inefficient distributor training. By leveraging Tencent Cloud Agent Development Platform, DaShenLin built the industry's first pharmaceutical AI knowledge base system, "AI Xiaoshen." This system effectively solves gaps between traditional LLMs and enterprise knowledge, addressing poor performance in specific domains and achieving millisecond-level precise responses. In sales scenarios, drug query efficiency improved by 80%; In office scenarios, cross-department collaboration was optimized; and decision-making is enhanced by analyzing feedback data from 600,000 users. Currently, AI Xiaoshen has been deployed across direct-operated and franchised stores nationwide. It continuously advances in all aspects, evolving from a knowledge Q&A assistant to a sales assistant and eventually to a decision engine, setting a new standard for AI-driven Q&A in the pharmaceutical retail industry.

## Strengths

1: Offers proprietary RAG algorithms and best practices tailored for enterprise services scenarios, along with various operational tools for generating and verifying Q&A pairs and conservative responses, ideal for scenarios like government services, policy consulting, and intelligent customer service.

2: Tencent Cloud Agent Development Platform provides advanced document parsing capabilities and mixed image-text layout capabilities for enterprises dealing with multimodal knowledge. It enhances the ability to process documents with mixed text, images, and tables, improves recognition accuracy based on OCR and LLMs, and effectively handles elements such as titles, formulas, headers, and footers. It is widely applicable for various industries, such as retail/travel manuals, and research reports, and more.

3: Tencent Cloud Agent Development Platform provides an innovative workflow solution for complex scenarios that require customized business processes, which enables the visual orchestration of atomic capabilities (including LLMs, knowledge bases, and plugins, etc.) via drag-and-drop. With clearly defined nodes, it supports zero-code creation of intricate workflows and handles multi-turn dialogue challenges, including multi-parameter extraction and rollback in different business scenarios. This makes it ideal for enterprise scenarios with complex processes, such as express delivery and registration booking, etc.

4: Tencent Cloud Agent Development Platform provides a multi-agent framework for scenarios that require more diverse tool integrations and more flexible conversational capabilities. It enables multi-agent collaboration through configurable delegation relationships and allows workflows to be executed in an agent-based manner, ensuring a structured yet flexible conversation flow. Additionally, the platform also supports the integration with a wide range of tools.

## Tencent Cloud TI Platform

### Challenges of enterprises

General LLMs are flourishing, but their lack of industry-specific knowledge and task-handling capabilities due to the absence of industry attributes hinders practical business deployment.

During the entire modeling lifecycle, algorithm engineers devote over 60% of their time to high-quality data preparation, including tasks such as data cleansing, labeling, and other preprocessing for large volumes of data.

In enterprise-grade model construction, stringent data access control is required, making end-to-end data security assurance highly challenging.

### Solution

Based on the internal LLM iteration experience of Tencent Cloud, the TI-ONE platform provides detailed algorithm task classification and intelligent data matching features, with over one million finely-tuned matching data entries. This covers over 100 LLM use cases across 12 major categories, addressing the issue of lacking industry-specific knowledge.

The TI-ONE platform provides a flexible data construction pipeline, with pre-configured, fully open-source, and extensible data preprocessing chain code. It supports three major model fine-tuning training scenarios: supervised training with multi-turn Q&A, supervised training with single-turn Q&A, and unsupervised pretraining. It covers five high-quality data processing steps: raw data analysis, data cleansing, data deduplication, prompt optimization, and training format conversion.

The platform provides flexible data schema configuration capabilities, automatically generating multiple LLM annotation consoles to enable a single annotation console to adapt to diverse labeling scenarios.

The platform leverages the permission system of Tencent Cloud to support directory-level access control for cloud storage used during the modeling process. It allows flexible configuration of resource isolation policies to meet enterprise-level data security control requirements.

## Use cases

The customer's core businesses include an e-commerce platform for the construction industry, construction labor outsourcing, and construction bidding services. The customer has an LLM algorithm team with strong capabilities in model training and development. The algorithm team fine-tuned the construction bidding LLM based on the TI platform to automatically assess bidding information and determine whether the company's capabilities match the bid requirements. In this industry, the bidding process often does not require the preparation of tender documents, relying instead on manual judgment. As a result, historical data only includes the tender requirements and the final bid outcome (win/loss), with no accumulated data on the intermediate bid-matching process. Therefore, high-quality training data needs to be constructed. The customer uses the online services of the TI platform to deploy the DeepSeek V3 model online, generating tens of millions of data pairs. The customer then uses the data processing tools of the platform to cleanse the data and continuously obtain high-quality bidding requirement matching data.

## Strengths

The platform provides a practical LLM fine-tuning and deployment toolchain, featuring the built-in Angel acceleration suite and based on YouTu algorithms. It offers industry-leading training and inference acceleration capabilities, along with efficient and flexible data preparation tools, stable and controllable task scheduling capabilities, and easy-to-maintain, high-performance service deployment capabilities. Stand-alone and multi-machine distributed service deployments can be initiated in just minutes with a few simple steps, significantly reducing efforts for Proof of Concept (PoC), production rollout, and maintenance.

# Data Platform solution

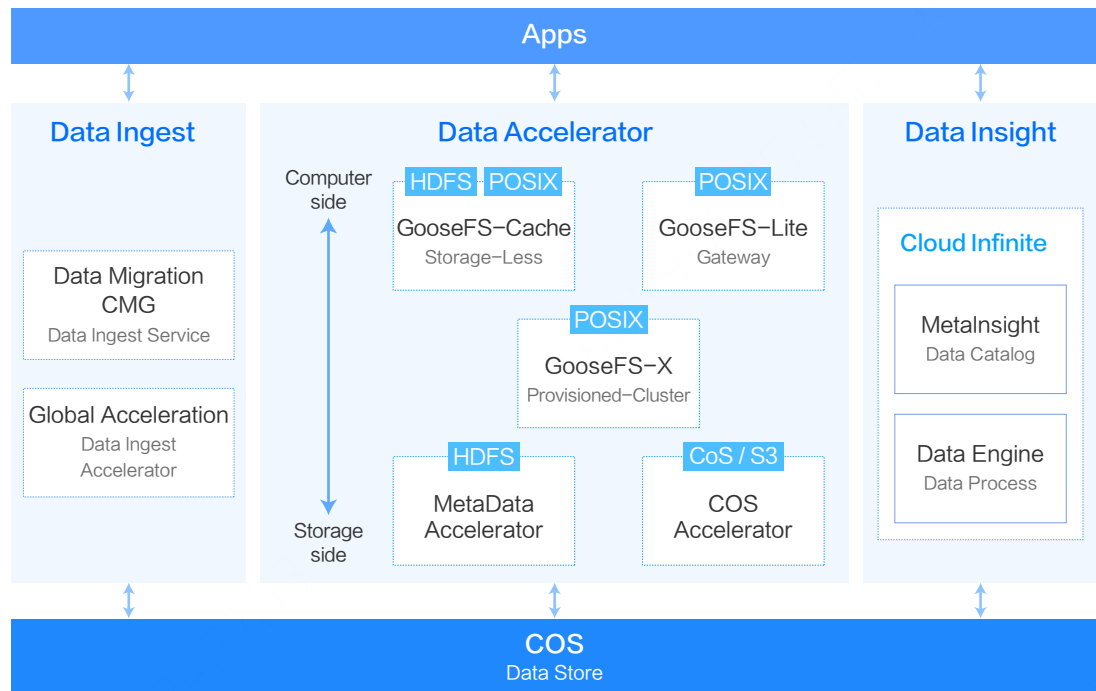
## Challenges of enterprises

Enterprise LLMs face challenges such as insufficient storage space, storage performance bottlenecks, and difficulties in data governance. The scale of data directly affects model accuracy. Massive amounts of data need to be rapidly collected from around the world and then aggregated into a unified storage pool. The mismatch between storage performance and computing power leads to inefficient data processing and model training. High-performance storage resources are required for model data preprocessing. In addition, cross-platform data flow and multi-protocol access must be supported to improve data processing and training efficiency. The complexity of data management and governance tasks severely affects operational efficiency, requiring the assurance of data quality, security, and compliance while also improving employee productivity.

## Solution

The Data Platform solution of Tencent Cloud consists of products such as Cloud Object Storage (COS), Data Accelerator Goose FileSystem (GooseFS), and Cloud Infinite (CI). It provides an end-to-end storage solution for data storage, training acceleration, and intelligent data management. The Data Platform solution provides enterprises with a high-performance, flexible, and scalable integrated data management solution. It provides strong technical support for enterprises to easily handle increasingly complex data processing needs while improving data efficiency and controlling storage costs.

Cloud Object Storage (COS) leverages Tencent Cloud's globally deployed acceleration nodes and data center (DC) access points to enable fast, easy, and large-scale data ingestion over the public network. As a centralized storage pool for massive data, COS provides reliable and secure storage capabilities, significantly improving system availability, reliability, and performance while reducing storage costs. GooseFS offers near-compute-layer data caching and acceleration services. Compatible with multiple data semantics, it integrates seamlessly with compute ecosystems to support the high-performance storage and checkpoint writing requirements of large-scale data processing and AI model training. CI provides near-storage-layer data governance capabilities. With intelligent search powered by Metalnsight, CI enables cross-modal retrieval of metadata and various media types. CI ensures both computational and content compliance while unlocking the full value of stored content. CI also reduces operational complexity and error risk, and improves the efficiency and security of end-to-end data management.



## Use cases

Tencent Cloud's Data Platform solution is deeply optimized for high-performance storage scenarios such as data lakes, artificial AI-generated content (AIGC), advanced driver-assistance systems (ADAS), embodied intelligence, and high-performance computing (HPC). A representative case involves a leading domestic LLM enterprise facing challenges in managing over 100 petabytes (PB) of multimodal data, complex processing workflows, and demanding governance requirements.

Tencent Cloud Object Storage (COS) leverages its global deployment and internal accelerated domain access to enable fast data transmission across all computing resources. Its features, such as automatic tiering and fragmentation cleanup, support full data lifecycle management, which further reduces storage costs.

GooseFS, equipped with a distributed metadata engine, delivers efficient metadata operations for files. It improves data cleansing in large-scale data lake environments.

CI, through its intelligent search engine Metalnsight, empowers teams to manage massive data sets with precision. It supports fine-grained, unified governance of hundreds of billions of data, which can help enterprises reduce operational costs while maximizing data value.

## Strengths



### High-speed public network access

Tencent Cloud offers stable and efficient data transmission services with 26 physical regions, over 3,200 global acceleration nodes, and dozens of DC access points. It supports terabyte-scale public network bandwidth, which ensures fast and reliable data ingestion for enterprise-scale model training.



### Massive-scale data storage

Powered by Tencent Cloud's fully self-developed object storage engine YottaStore, COS delivers a secure and reliable distributed storage platform. A single COS cluster can manage data at the exabyte scale to meet the demands of persistent data storage in the AI era.



### High-performance data caching

GooseFS supports throughput at the terabits-per-second (Tbit/s) level, with the ability to handle hundreds of billions of metadata entries. Single-client link speeds can easily reach gigabytes per second (GB/s), which delivers a tenfold performance boost over the industry norm and significantly accelerates LLM distribution.



### Multi-protocol access

GooseFS connects to a wide range of data sources and provides upper-layer computing applications a unified namespace with support for multiple access protocols such as POSIX, Hadoop Distributed File System (HDFS), Kubernetes Container Storage Interface (CSI), Filesystem in Userspace (FUSE), and COS semantics for seamless data flow across diverse storage systems.



### End-to-end data processing

CI offers over 100 built-in data processing capabilities and enables automated processing of data stored in COS. With intelligent search powered by MetalInsight, CI enhances the efficiency of data processing, management, analysis, and retrieval.

## Data Engine solution

### Challenges of enterprises



### Traditional storage cannot meet the performance demands of modern AI workloads

With the rise of multimodal training and end-to-end vision-language-action (VLA) models, storage systems face significantly increased read/write performance requirements for higher throughput, lower latency, and greater input/output operations per second (IOPS).



### Data silos across multi-cloud environments

Most users use multiple public cloud providers along with storage from their own Internet data centers (IDCs). Connecting isolated data silos across these platforms to enable efficient data mobility has become a critical challenge.



### Diverse API requirements exceed the capabilities of single storage systems

AI workflows across stages, such as data cleansing, training, inference, and simulation, require different APIs. Standard storage systems cannot support multiple APIs simultaneously, which leads to repeated data copying and reduced efficiency.



### Managing data at the hundred-billion scale poses serious challenges

When data volume reaches the hundred-billion level, the lack of efficient data management tools prevents quick access to information such as data temperature and directory size. This leads to rising costs and loss of control over storage capacity.

## Solution

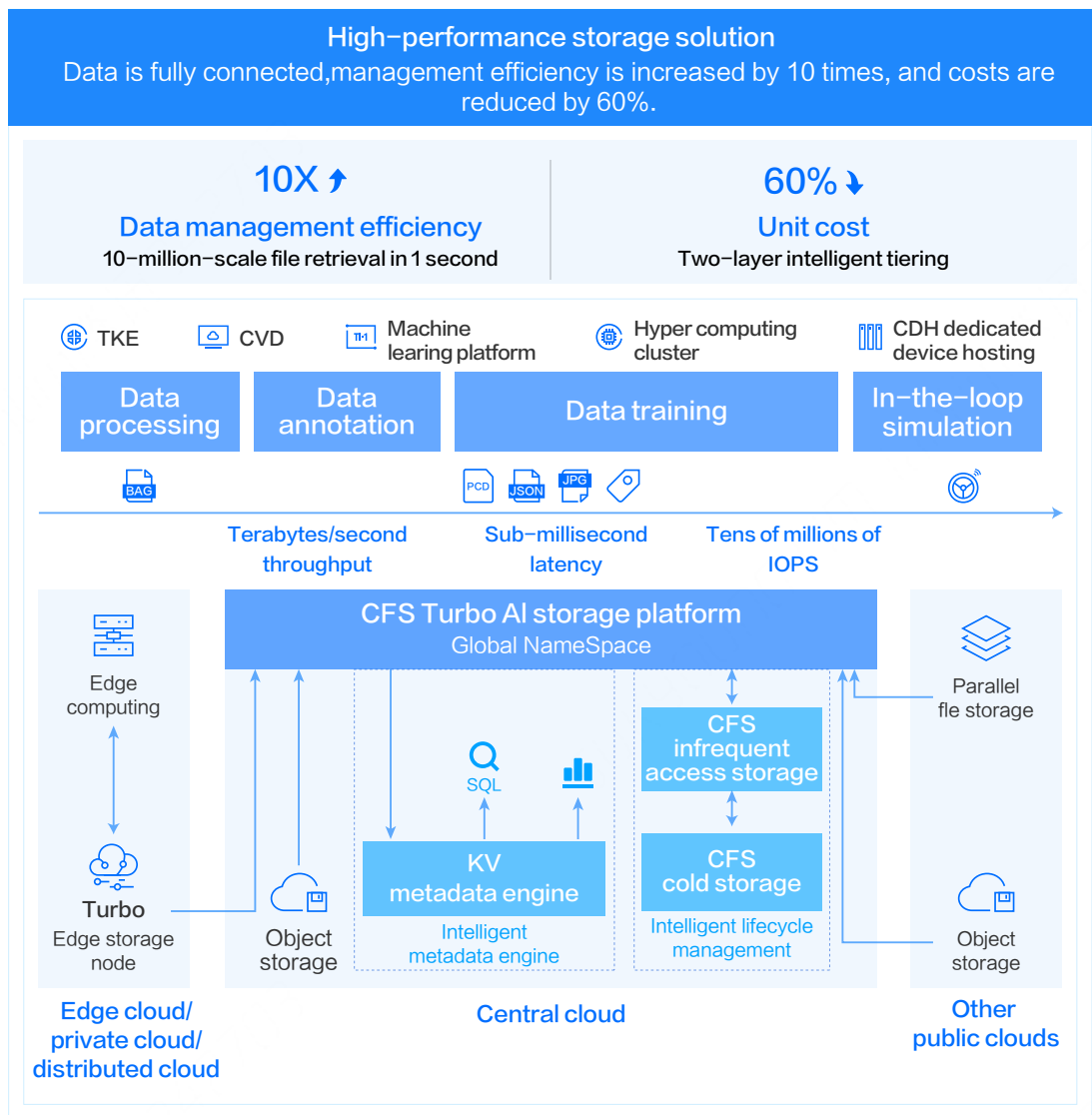
Tencent Cloud offers an AI storage platform built around the high-performance file storage service CFS Turbo. This one-stop solution addresses high-performance storage requirements in scenarios such as autonomous driving, LLM training, and HPC, through a simplified architecture.

In terms of performance, the platform adopts an end-to-end parallel processing architecture and leverages kernel-mode clients, Remote Direct Memory Access (RDMA), and GPU Direct Storage to significantly shorten the I/O path. It delivers terabytes-per-second (TB/s) throughput, tens of millions of IOPS, and sub-millisecond latency. This meets the demanding performance needs of AI scenarios, including fast read/write of large checkpoint files, concurrent access to hundreds of billions of training samples, and rapid loading of terabyte-scale model files. It greatly improves training and inference efficiency in AI workloads.

Secondly, with metadata mapping capabilities, the platform enables seamless integration across public cloud object storage, public cloud parallel file storage, and on-premises storage. It creates a globally unified namespace, which significantly improves data mobility and meets the requirements for efficient data flow in complex multi-cloud environments.

Thirdly, in addition to full POSIX semantics, the platform also supports HDFS semantics, meeting diverse API requirements across various stages such as data cleansing, labeling, training, and inference. It seamlessly integrates with cloud-based services, including container platforms, ML platforms, Cloud Virtual Desktop (CVD), Elastic MapReduce (EMR) big data platforms, and Hyper Computing Cluster (HCC), fully addressing storage needs across all scenarios.

Lastly, the platform offers two key differentiated capabilities for large-scale data storage scenarios to enhance the user experience. First, the intelligent metadata engine completes the retrieval of tens of millions of files within seconds, increases data management efficiency by a factor of ten, and enables efficient management of data at the hundred-billion scale. Second, the intelligent data tiering mechanism applies storage policies to automatically implement hot-cold data separation without modifying business applications, reducing costs by up to 60%.



## Use cases

### Training scenarios

LLM user A adopted Tencent Cloud CFS Turbo and achieved over 20 GB/s write bandwidth with a 100 TB capacity, along with 100,000-level deletion operations per second. This resolved challenges such as burst high-bandwidth writes, small-capacity yet high-bandwidth loads, and massive file deletions during GLM-130B LLM training, significantly improving training efficiency.

LLM user B adopted Tencent Cloud CFS Turbo to support PB-scale metaverse rendering and LLM training. By using elastic container services, the user achieved scaling to thousands of nodes within minutes and reached 70 GB/s of high-concurrency read/write bandwidth, accelerating AI training and visual rendering with ultra-low latency of 3.5 ms.

LLM user C used Tencent Cloud CFS Turbo to sustain high-frequency TB-scale checkpoint writes every 2 hours at a peak bandwidth of 108 GB/s with 100% availability. Combined with elastic scaling, the solution ensured continuous and efficient training iterations and sped up the foundation LLM development for language AI.

### Inference scenario

User D deployed inference services with CFS Turbo and completed loading the deepseek 671 B model in 5 minutes. The single-thread loading speed reached 1.5 GB/s, a sixfold improvement, greatly enhancing the efficiency of spinning up GPU instances.

### Full AI workflow scenario

User F adopted Tencent Cloud's Data Engine solution to connect with other public cloud storage systems under a unified namespace. The user accessed HDFS for data cleansing and used POSIX semantics for training and inference, which enabled zero-copy data access and significantly improved efficiency. With intelligent data lifecycle management and a smart metadata engine, the user efficiently managed 23 billion data records, reduced costs by 65%, and improved analytics efficiency tenfold.

## Strengths



### Exceptional performance

CFS Turbo delivers throughput at the TB/s level, tens of millions of IOPS, and millions of metadata operations per second (OPS). It completes checkpoint file read/write within 10 seconds and handles tens of millions of file read/write operations within 1 second. Single request latency is in the hundred-microsecond range, which meets the sample reading demands of large-scale clusters with tens of thousands of GPUs for LLM training.



### AI storage with platform-level capabilities

CFS Turbo features a streamlined architecture and stands as the only parallel file storage service among cloud providers that supports multiple protocols, including POSIX and HDFS semantics. It seamlessly integrates with public cloud object storage, public cloud parallel file storage, and on-premises storage across diverse scenarios and data types. This creates a globally unified namespace, offering an all-in-one storage solution that significantly improves data processing efficiency for AI workloads.



### Rich functionality

CFS Turbo is the only parallel file storage service among cloud vendors to support automated data tiering between hot and cold storage, so as to help users save over 60% in storage costs. It also provides advanced features such as snapshots, encryption via Key Management Service (KMS), metadata engine, directory quotas, user quotas, and audit logs, meeting sophisticated management needs across user levels.

On the Data + AI platform, CI and the log management service empower users with deep metadata insights and flexible data analysis. By leveraging AI to continuously uncover data patterns, the platform helps transform data into real business value.

## CI MetaInsight

### Challenges of enterprises



### High data governance costs

Unstructured data comes from an increasing number of sources and channels, and its volume grows exponentially. Traditional governance methods rely on manually maintained documentation, which is time-consuming and quickly becomes outdated. Without a unified tool for management and retrieval, businesses struggle to reduce data governance costs and locate the right files efficiently.



### Limited data value exploration

Massive volumes of data lack proper classification, correlation, and reuse. This leads to repeated development of data management tools, inefficient use of data, and missed opportunities to unlock its full value.





### Restrictions of single-modality search

Traditional search methods remain limited to a single modality, such as text-to-text search. They fail to support cross-modal data correlation and rely solely on keyword matching, which makes it difficult to capture deep semantic relationships and enable natural language search.

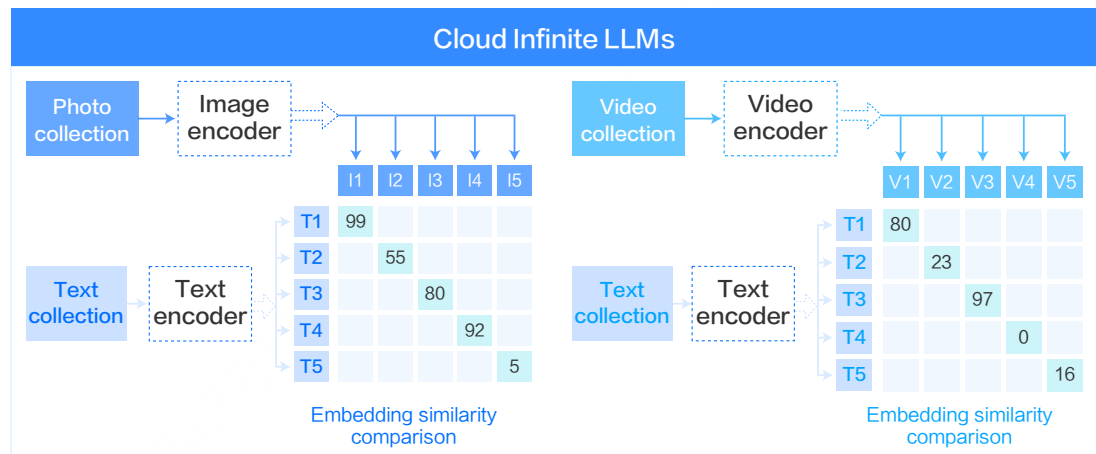
## Solution

To address the challenges faced by enterprise customers in the AI era, such as exponential growth in data scale and file volume, inefficient search, and low metadata utilization, CI integrates a self-developed meta-data management platform to launch an intelligent data search product. This product supports hybrid search of vector and structured tags across all media types. On the Data + AI platform, MetaInsight provides multi-scenario, cross-modal search services that help customers manage, analyze, and retrieve stored data effectively.



### Cross-modal search service

Supports all media file types by extracting multi-modal features from massive Chinese language corpora and mapping them into a unified feature space to enable cross-modal associative search.



### Efficient massive data analysis

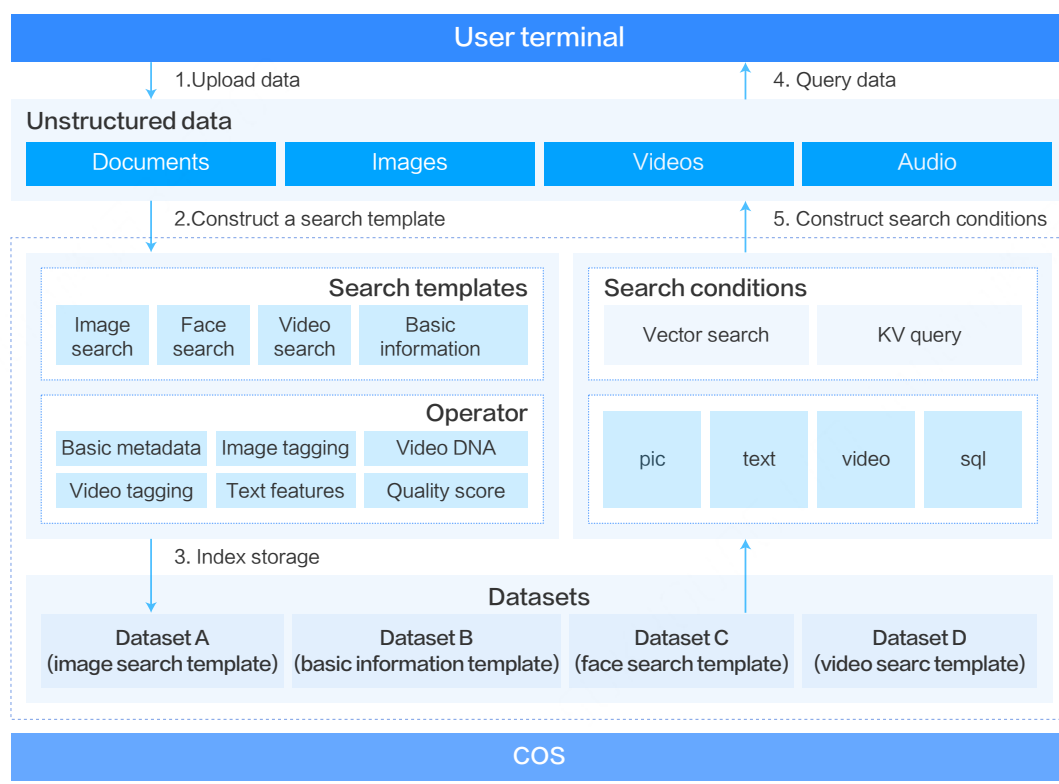
Enables fast and precise queries across billions of files for specific criteria, such as locating a particular file, counting a batch of files, or calculating the proportion of image formats.



### One-stop search platform

Allows keyword-based file queries, such as finding files containing specific keywords or videos featuring certain individuals, facilitating better data management, analysis, and value extraction.

## MetalInsight



## Use cases

### Scenario 1: E-commerce product search

MetalInsight enables image-based product search and allows users to upload a photo and instantly find the same or similar items. By matching visual features instead of relying on keywords, it solves the common challenge of inaccurate search results in traditional e-commerce platforms.

### Scenario 2: smart photo albums

With facial recognition powered by MetalInsight, users can easily locate photos featuring the same person. Images are ranked by similarity, and with support for hundreds of millions of metadata management, the system delivers near-instant search results, even across billions of photos, moving beyond basic filters like time or location.

### Scenario 3: data pre-sorting for AI training

Before training AI models, MetalInsight helps organize massive unstructured data sets through intelligent pre-sorting. For example, by entering descriptions like "rainy day" or "pedestrian crossing", users can quickly pull relevant data. Compared to manual labeling, this solution reduces preprocessing time by over 70%. It also supports multimodal search, such as combining image and text inputs, to help enterprises efficiently build high-quality training data sets during the data cleansing stage.

## Strengths



### Multimodal search

supports various search methods, including text-to-image and image-to-image queries, along with efficient metadata query and statistical analysis.



### Comprehensive coverage

Covers thousands of industry-specific scenarios. Users can choose from built-in templates tailored to their needs and quickly build search applications to maximize content value.



### High performance

Enables queries on hundreds of billions of data items with millisecond-level latency and over 95% recall accuracy.

## Cloud Log Service (CLS)

### Challenges of enterprises

As businesses accelerate their digital transformation, IT systems are becoming increasingly complex. Such a trend poses serious challenges to traditional Ops and maintenance:



### Log analysis is hard to start

Logs are often huge in volume, inconsistent in format, and difficult to interpret. Manual review is time-consuming, relies heavily on experience, and comes with a steep learning curve.



### Inefficient root cause analysis

Diagnosing issues often requires correlating logs, metrics, and traces across multiple sources. Manual investigation is time-intensive and relies heavily on expert knowledge, and it is hard to standardize.



### Reactive instead of proactive

Most anomaly detection relies on pre-set rules and expert input, which makes it hard to catch unknown issues early. Proactive alerts are often lacking.



### Tools don't work well together

Data is scattered across monitoring systems, databases, Configuration Management Database (CMDB), and third-party platforms. The lack of integration leads to siloed workflows and high collaboration costs, and it is difficult to close the loop.

### Solution

Traditional algorithm-based AIOps platforms often rely heavily on consistent monitoring data sources and well-defined application system topologies. This limits their flexibility and makes it difficult to support diverse and evolving business system architectures.

In the Data + AI platform, Agentic Analytics significantly enhances the value of data-driven business by using intelligent data management, automated decision-making, and proactive analysis capabilities. CLS, enhanced by LLM technology, integrates deeply with both data and AI capabilities to build an intelligent data hub that delivers an end-to-end MLOps solution.

## Core capabilities



### Multimodal data processing

supports unified ingestion of logs, metrics, and tracing data. Compatible with MCP protocols and third-party data sources such as databases and CMDBs.



### LLM-powered analysis

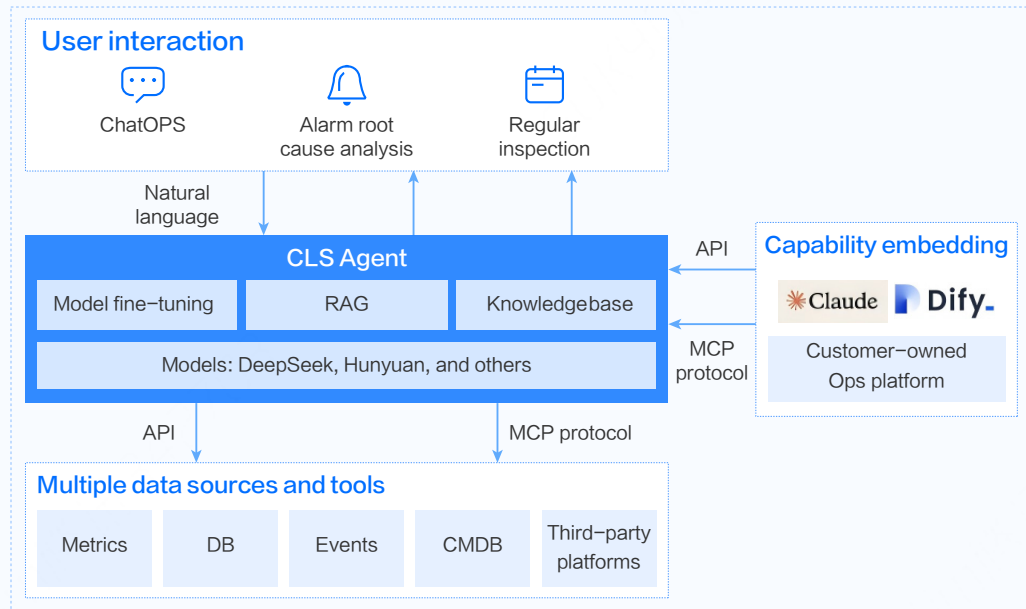
leverages domain-specific fine-tuned LLMs to autonomously plan Ops troubleshooting workflows, query monitoring data, and perform automated root cause analysis based on an operations knowledge base.



### Dynamic decision engine

Evolved from rule- and algorithm-driven logic to a dual-engine approach powered by real-time data and a knowledge base. Enables real-time anomaly detection, risk simulations, and automated remediation.

## Technical architecture



## Use cases

### Scenario 1: intelligent alarm and root cause analysis

Integrate logs and Tracing data through CLS, automatically connect error logs with service call chains by using LLM, and reduce time for root cause analysis from hours to minutes.

### Scenario 2: automated log parsing

Automatically identify unstructured raw logs and dynamically generate regular expressions and index configurations to structure logs and improve subsequent query efficiency.

### Scenario 3: natural language interactive Ops

Trigger intelligent inspections or abnormal analysis via natural language prompt. automatically generate visual charts and analytical reports through CLS, and reduce configuration time from 30 minutes to 1 minute.

## Strengths



### LLM fine-tuning for vertical fields

refers to a specialized model trained based on Tencent's extensive Ops scenario data, which achieves over 90% accuracy in querying monitoring data and provides more efficient self-optimized troubleshooting workflows.



### Full stack data fusion capability

supports unified analysis of multimodel data, including logs, metrics, and Tracing data, and natively integrates monitoring data from commonly used cloud services.



### Barrier-free interaction design

supports natural language interaction throughout data queries, alarms, and dashboard configuration, which simplifies AI technology adoption.



### Enterprise-grade closed-loop capability

provides full lifecycle management from risk alerts (inspections) to incident troubleshooting (root cause analysis) and post-event optimization (knowledge accumulation), which shifts Ops features from response to prevention.

## Tencent Cloud data and intelligent security

### Challenges of enterprises

As enterprises adopt LLM and AI, both business and IT face significant security challenges: Data serves as the "fuel" for AI, but sensitive data is highly prone to leakage during training and inference, such as via prompt word injection or insecure API interfaces. The "black box" nature of AI models complicates its decision-making traceability. Emerging AI attack methods, like prompt word injection, may lead to compliance risks and business losses. As a "universal connector" in the agent ecosystem, the MCP protocol also brings about a series of security issues, such as protocol fragmentation, traditional application access security control, identity authentication, and unified access control. The autonomous behavior of AI agents expands the attack surface, and problems such as credential management and unauthorized access become prominent, posing a severe test to the existing security system. There is an urgent need for trustworthy, controllable, and reliable AI security protection.

### Solution

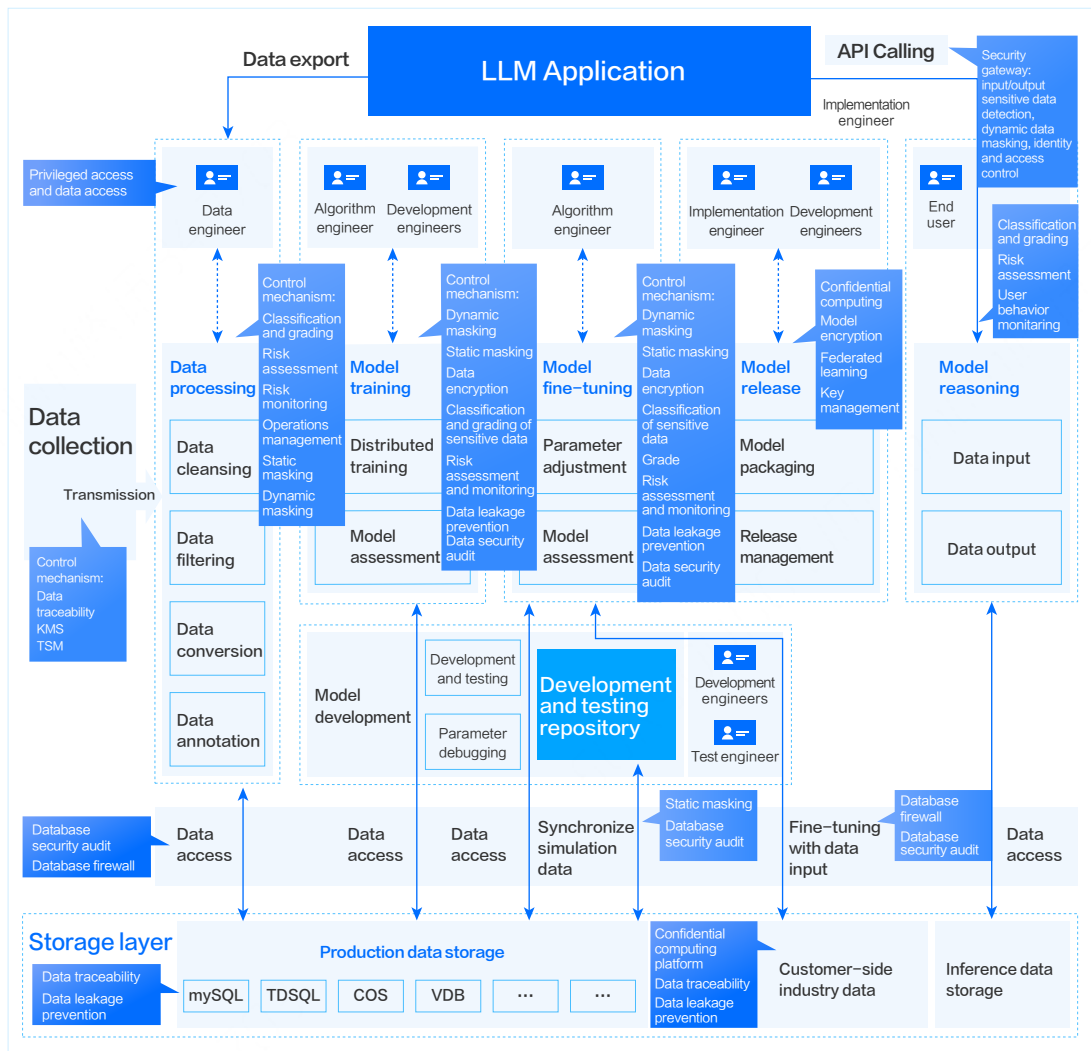
To address security challenges posed by LLM and AI, enterprises must establish an integrated "Data + AI" security framework. This requires deep convergence of data security governance and AI lifecycle security to enable end-to-end risk control.



### LLM data security protection

**Data classification, grading, and masking:** Identify, classify, and grade sensitive data such as PII/PHI and business secrets during data access and processing stages, minimize data leakage risks while preserving data usability by using techniques such as SM encryption, masking, and data anonymization. Especially when building a RAG knowledge base or fine-tuning a model, it is important to ensure the compliance of the input data.

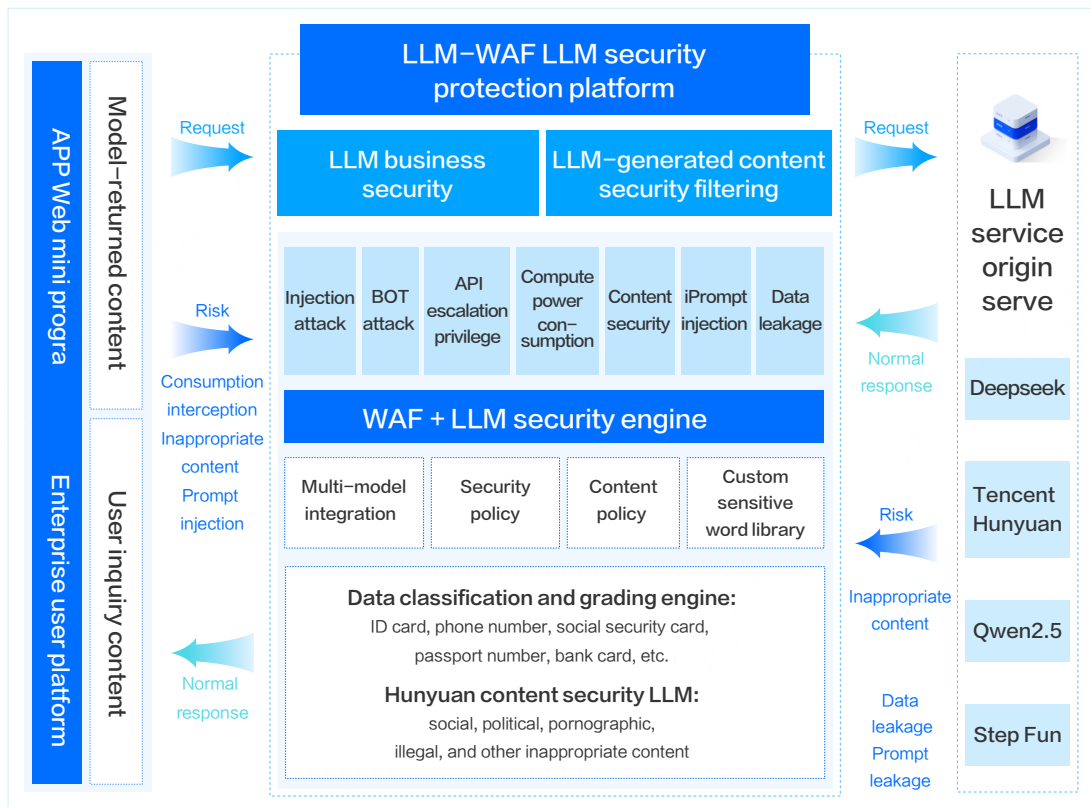
**Secure data flow and storage:** Establish security measures covering the entire data lifecycle, including transport encryption, storage encryption, and access control, and establish specialized security policies for emerging storage solutions like Tencent Cloud VectorDB.



### Protection of AI infrastructure and running environment

AI Large Language Model Web Application Firewall (LLM-WAF) establishes comprehensive application and API protection capabilities at the boundary of intelligent applications. It detects malicious commands in user inputs (prompts), filters sensitive information, and conducts compliance reviews of model outputs. This enables end-to-end security across multi-model, multi-scenario, and high-concurrency environments. LLM-WAF also supports real-time detection and interception of computing power abuse, prompt word attacks, and data leakage risks targeting LLMs, helping enterprises build a reliable, stable, and sustainable LLM service ecosystem.

AI-SPM is a security management system that protects LLM infrastructure and running environment and detects LLM attack surfaces and vulnerabilities. It detects and handles security risks in a timely manner through external attack surface detection, AI infrastructure software component management, and combined threat intelligence and vulnerability detection.



### AI model and application security protection

T-Sec LLM security gateway, serving as the intelligent security hub for enterprise AI application deployment, connects agents, models, and services to enable unified governance and efficient collaboration. It addresses critical risks in AI scaling through a multi-layered protection mechanism.

**Identity and permission management:** Assign independent and least-privilege credentials to AI agent while avoiding hard-coded credentials and establish strict access control policies to limit the operational scope and data access permissions.

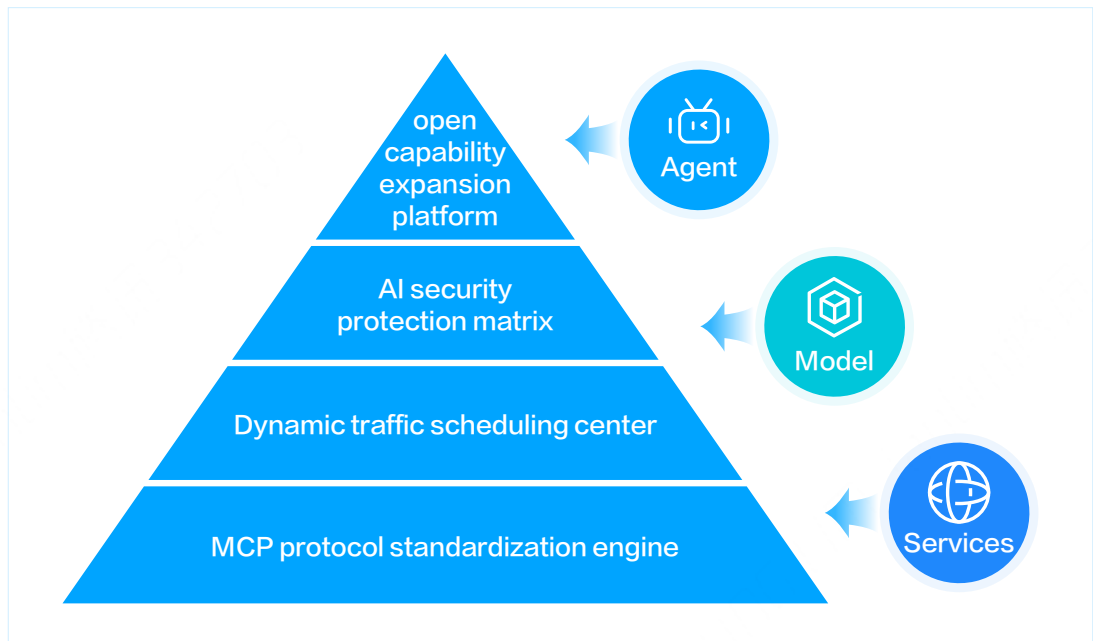
**Behavior monitoring and auditing:** Perform real-time monitoring, logging, and auditing of AI agent activities to promptly detect abnormal behaviors and potential threats such as unauthorized access or data breaches.

**Full-linkage security management:** It supports accurate identification and handling of various security risks in LLM scenarios such as input prompts, content generation, and running environment, including malicious instruction injection, tool poisoning attacks, and access to sensitive information. It also has flexible control granularity and supports policy configuration based on MCP server and MCP tools dimensions, providing users with rich and comprehensive security protection capabilities.

**MCP protocol encapsulation:** Provide standardized protocol conversion capabilities, support one-click encapsulation of traditional API into standardized MCP services, and simultaneously generate Client SDK to achieve seamless integration of agent, effectively lowering the development threshold and helping enterprises quickly integrate existing workloads into the MCP ecosystem.

**Gateway traffic control:** Enable MCP Server-based routing, forward/reverse proxying, and full-dimensional traffic management. Dynamically route traffic across versions, models, and tools based on parameters, protocols, and load metrics.

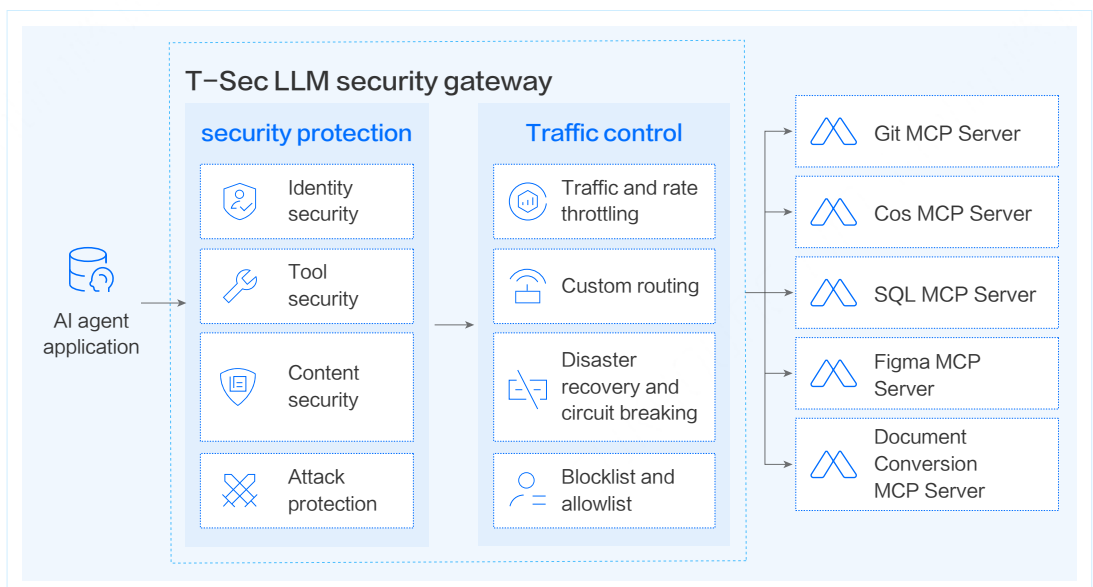
**Multi-dimension protection system:** Based on an open security capability platform, it provides standardized interfaces for the rapid integration of third-party service components such as content security and data security. Users can flexibly configure scenario-based protection strategies such as text/image multimodal content review and structured data security detection to build a comprehensive LLM security governance system.



## Use cases

### Scenario 1: security protection of enterprise agent applications

A manufacturing enterprise employs AI agents to monitor production line data and automatically control equipment parameters. LLMs face multiple security threats during large-scale deployment, such as injection attacks, sensitive data leakage, and MCP Tools poisoning. Enterprises need to build an intelligent security protection system that covers the entire linkage from input to output. By deploying the T-Sec LLM security gateway, we establish multi-dimension attack defense for LLMs, identify and intercept attack behaviors such as prompt injection, unauthorized tool access, abnormal traffic, and have built-in sensitive data identification rules to desensitize high-risk fields such as ID card numbers and phone numbers to protect enterprise systems and information security. The system monitors agent operation prompts and data access behaviors in real time. Upon detecting any attempt to access unauthorized data or perform unauthorized actions, it immediately triggers alarms and blocks the operation to prevent production incidents or data breaches caused by malicious exploitation of agents.





### Scenario 2: LLM security protection for retail enterprise intelligent customer service system

A certain retail chain has launched an intelligent customer service system based on a LLM to provide consumers with order consultation, return and exchange policy answers, and product recommendation services. To prevent personal sensitive information from being included in user input and LLM output, or malicious users from launching prompt injection and induced attacks, and the model from being misled to generate illegal, sensitive, or inappropriate content, the company connected to Tencent Cloud LLM-WAF solution. During the input stage, the system can identify private data such as mobile phone numbers and addresses in real time and prompt users, while detecting and blocking potential injection attacks and prompt word-induced behaviors; during the output stage, the results generated by the LLM are subject to content review to prevent the leakage of personal sensitive information or the output of non-compliant statements. The system also supports log retention and policy optimization of risk interaction content, helping companies to improve the intelligent customer service experience while ensuring compliant operations and brand reputation security.

At the same time, by deploying AI-SPM, comprehensive risk management and security monitoring are carried out for the enterprise's LLM infrastructure and intelligent application production and running environment. Combined with external exposure management, threat intelligence, and vulnerability detection, the security status of AI applications and infrastructure is comprehensively evaluated and operated, completing the management and response closed loop to ensure safe and stable workload operation.

## Strengths

Tencent Cloud showcases exceptional expertise in the "Data + AI" security domain, providing enterprises with robust safeguards for developing secure and trustworthy AI applications:



### Full-stack security capabilities and expertise

Tencent Cloud offers a comprehensive security product matrix spanning infrastructure security (IaaS), platform security (PaaS), and application security (SaaS), coupled with proven intrusion prevention experience. This enables multi-layered and in-depth protection for AI systems.



### Leading data security governance practices

Tencent possesses extensive expertise in massive-scale data processing and security governance. Its data security product suite, including Tencent Cloud Data Security Governance Center (DSGC) and Database Audit, delivers comprehensive protection for AI training and inference data throughout the entire lifecycle.



### Robust AI technology accumulation and ecosystem

Tencent integrates proprietary AI technologies like Tencent Hunyuan LLM with industry-specific solutions to deliver business-aware AI security capabilities such as targeted content moderation and AI agent behavior analytics.



### Cloud-native and open integration

Tencent Cloud platform provides cloud-native security services designed for effortless integration and scalability, enabling seamless connectivity with existing enterprise IT architectures and third-party security tools. This empowers enterprises to rapidly build tailored "Data + AI" security systems that meet their specific requirements.

Leveraging these strengths, Tencent Cloud is committed to empowering enterprises to fully harness the benefits of AI technology while effectively managing and mitigating potential security risks, thereby achieving sustainable intelligent transformation.

# 05.

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## Industry-specific use cases of Data + AI

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The concept of Data + AI has become highly prevalent, demonstrating extensive application scenarios across nearly all industries and sectors. The application scenarios of Data + AI in key sectors are as follows:

#### **In the financial sector**

LLMs are used to analyze real-time trading data, monitor market dynamics, adjust investment strategies, enhance returns, and reduce risks. They can provide precise risk assessments and fraud detection for financial institutions based on data such as customer credit histories, behavioral characteristics, and social relationships. Moreover, AI can work with technologies such as graph neural network, generative adversarial network, and anomaly detection to uncover latent risk factors, identify abnormal behaviors, and protect against financial losses. Based on user risk preferences, market trends, and historical return data, AI algorithms generate dynamic investment portfolios and use natural language generation (NLG) technology to output interpretable reports. In addition, predictive models trained on real-time news and social media sentiment data can anticipate market fluctuations.

#### **In the manufacturing sector**

sensors are utilized at every stage of production to collect real-time data on machine conditions, production speed, and material consumption. Such data can be thoroughly analyzed by the AI system to promptly identify production bottlenecks and issues such as declining machine efficiency or prolonged stages, and provide appropriate optimization solutions. In the supply chain sector, LLMs can be used for real-time data monitoring across all supply chain management stages, such as raw material supply, logistics, and inventory management. This improves resource allocation, boosts operational efficiency, and reduces costs. Additionally, AI algorithms can generate optimal transportation routes based on real-time traffic and weather data.

#### **In the healthcare sector**

LLMs can analyze vast biomedical data sets to automatically identify target genes, investigate biochemical and pharmacological mechanisms, assist researchers in understanding disease heterogeneity, and pave the way for future target discovery. In medical diagnosis, imaging data can provide visual information of lesions, while pathology data can provide detailed information at the cellular level. Combining these multimodal data can provide a more comprehensive understanding of the condition and provide comprehensive diagnostic recommendations, thereby improving the accuracy of diagnosis. By processing physiological data (such as blood glucose and heart rate) from smart wearable devices, AI models can provide early warnings for abnormalities and propose intervention measures accordingly. Combining public health data with climate information facilitates the modeling of infectious disease transmission. During the COVID-19 pandemic, similar methods were utilized to assess regional infection risks.

#### **In the retail sector**

AI enhances demand forecasting accuracy by analyzing historical sales data, market trends, and external factors (such as seasons and promotional events). For example, an AI+BI integrated system can accurately forecast SKU shipments and predict distributor POs, thereby improving prediction accuracy and optimizing full-link supply chain efficiency. GenAI can also simulate product sales trends in diverse market scenarios, aiding inventory policy development and minimizing the risks of oversupply or stockouts. AI-powered product recommendation systems leverage user behavior data (such as search histories and purchase records) to create personalized marketing content. For example, AI search tools enhance the shopping experience by enabling users to find products through contextual scenarios rather than just product names.

### **In more complex assisted driving scenarios**

real-time environmental data is collected via multi-modal sensors (including LiDAR, cameras, and millimeter-wave radars) and is combined with GPS/IMU to achieve centimeter-level positioning accuracy. AI processes and integrates multi-modal data, including visual images, radar signals, and GPS coordinates, to enable precise environmental perception and decision-making. AI models (utilizing reinforcement learning and game theory algorithms) analyze real-time traffic flows, obstacle trajectories, historical driving data, and other inputs to generate optimal route plans and predict the intentions of surrounding vehicles and pedestrians. This enhances driving safety by enabling quick responses to unexpected events. Training with synthetic data can boost the model's generalization capabilities, especially in edge scenarios such as blind-zone intrusions.

Furthermore, as AI technologies advance and data volumes grow exponentially, the application scenarios for Data + AI will expand and deepen, delivering transformative impacts and enhanced efficiency across industries.

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## Future of Data + AI

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Gartner forecasts that by 2028, 80% of GenAI business applications will be developed on organizations' existing data management platforms, reducing implementation complexity and time to delivery by 50%. By 2028, data sovereignty and bias for AI models to regulations from at least three geographies will account for 50% of data management workloads for AI. By 2027, 60% of AI tool usage within organizations will be unsupervised, prompting increases in investment in active metadata solutions to track use.<sup>6</sup>

In the future, the integrated Data + AI platform will bring AI closer to data, driving a leap forward in enterprise AI development. Technical advancements are particularly critical in six key areas, including the convergence of data management ecosystems, the application of natural language interactions, and the enhancement of data quality and governance.

## **Convergence of data management ecosystems**

The previously fragmented data management markets will be converged into a unified market centered on data ecosystems (powered by data fabrics and GenAI), which significantly reduces technical complexity and integration costs. Key initiatives involve stopping the deployment of siloed solutions, adopting integrated data ecosystems, and investing in metadata management and data fabric technologies.

## **Natural language interaction popularization**

In the future, natural languages will become the primary means for querying and operating data management ecosystems, greatly improving data consumption efficiency. Non-technical users can directly access data using natural languages, minimizing dependence on traditional semantics layers and modeling tools while fostering data democratization.

## **GenAI integration with existing platforms**

Enterprises can develop GenAI applications on their existing data management platforms such as data lakehouses. Such resource integration reduces implementation complexity and delivery cycles. Technically, retrieval-augmented generation (RAG) can be adopted to use existing platforms as knowledge bases, thereby enhancing application stability and ensuring data compatibility.

## **Data quality and governance enhancement**

Many GenAI projects are abandoned during the proof-of-concept phase due to poor data quality and insufficient risk control. Enterprises can improve the accuracy of code/query generation by using active metadata management, prevent AI hallucinations and privacy leaks, and use synthetic data to enrich training datasets.

## **Rise of Agentic AI**

Agentic AI will revolutionize software interfaces by replacing traditional GUIs with multimodal interactions, such as natural languages and gestures, while transforming end-to-end workflows into collaborative tasks involving multiple agents. Enterprises need to establish an adaptive governance framework to ensure the security and verifiability of autonomous AI systems.

## **Improvements in data infrastructure adaptability and compliance**

Enterprises will increasingly prioritize direct ROI from data infrastructure. For example, enterprises can collaborate with cloud-neutral vendors to improve cross-cloud data integration efficiency and enhance data readiness for AI, such as metadata management and AI vendor collaboration.

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<sup>6</sup>Data Management Is the Sole Differentiator in a Commoditized and Multipolar LLM World, G00828308

## Summary

Data + AI will not only transform the way enterprises operate but also create new sources of growth. Only by elevating Data + AI to a strategic core can enterprises realize its value and accelerate intelligent transformation to sustain competitiveness and lead the market. Looking ahead, advancements in multimodal LLMs will drive enterprise data platforms to focus on real-time performance, self-adaptive capabilities, and ethical compliance, forming the core infrastructures for GenAI deployment within enterprises.

